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Brown et al.

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(54) **MULTI-SOURCE SMART-HOME DEVICE CONTROL**

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(73) Assignee: **Google LLC**, Mountain View, CA (US)

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H04L 12/28 (2006.01)
G06F 16/2457 (2019.01)

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CPC ***H04L 12/2814*** (2013.01); ***G06F 16/24573*** (2019.01); ***H04L 12/2809*** (2013.01); ***H04L 12/282*** (2013.01); *H04L 2012/2847* (2013.01)

(58) **Field of Classification Search**
CPC H04L 12/2814; H04L 12/2809; H04L 12/282; H04L 2012/2847; H04L 12/2807; G06F 16/24573
USPC 709/223
See application file for complete search history.

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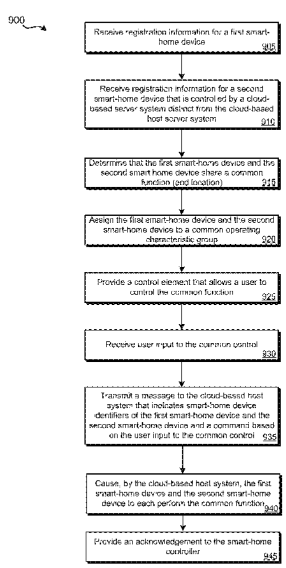
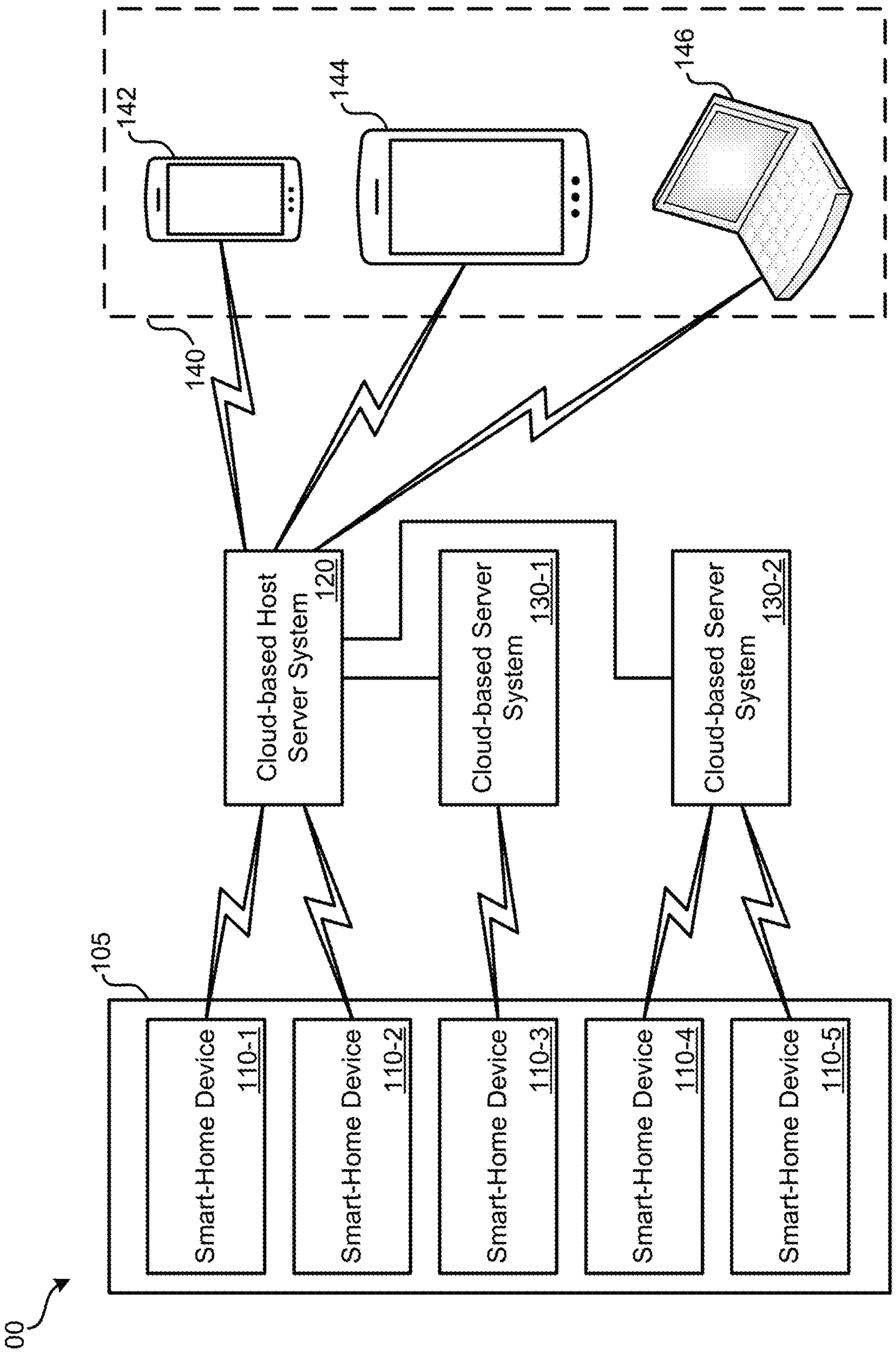
(57) **ABSTRACT**

Various arrangements for integrating control of multiple cloud-based smart-home devices are presented. Registration information may be received for a first and second smart-home device that are controlled using different cloud-based server systems. A determination may be made that that the first smart-home device and the second smart-home device share a common function. The first smart-home device and the second smart-home device may be assigned to a common operating characteristic group based on the common function being shared by the first smart-home device and the second smart-home device. A control element may be provided that allows for control of smart-home devices with the common operating characteristic group. The control element may be used to control the common function.

19 Claims, 15 Drawing Sheets





100
105
Smart-Home Device 110-1
Smart-Home Device 110-2
Smart-Home Device 110-3
Smart-Home Device 110-4
Smart-Home Device 110-5
Cloud-based Host Server System 120
Cloud-based Server System 130-1
Cloud-based Server System 130-2
140
142
144
146

FIG. 1

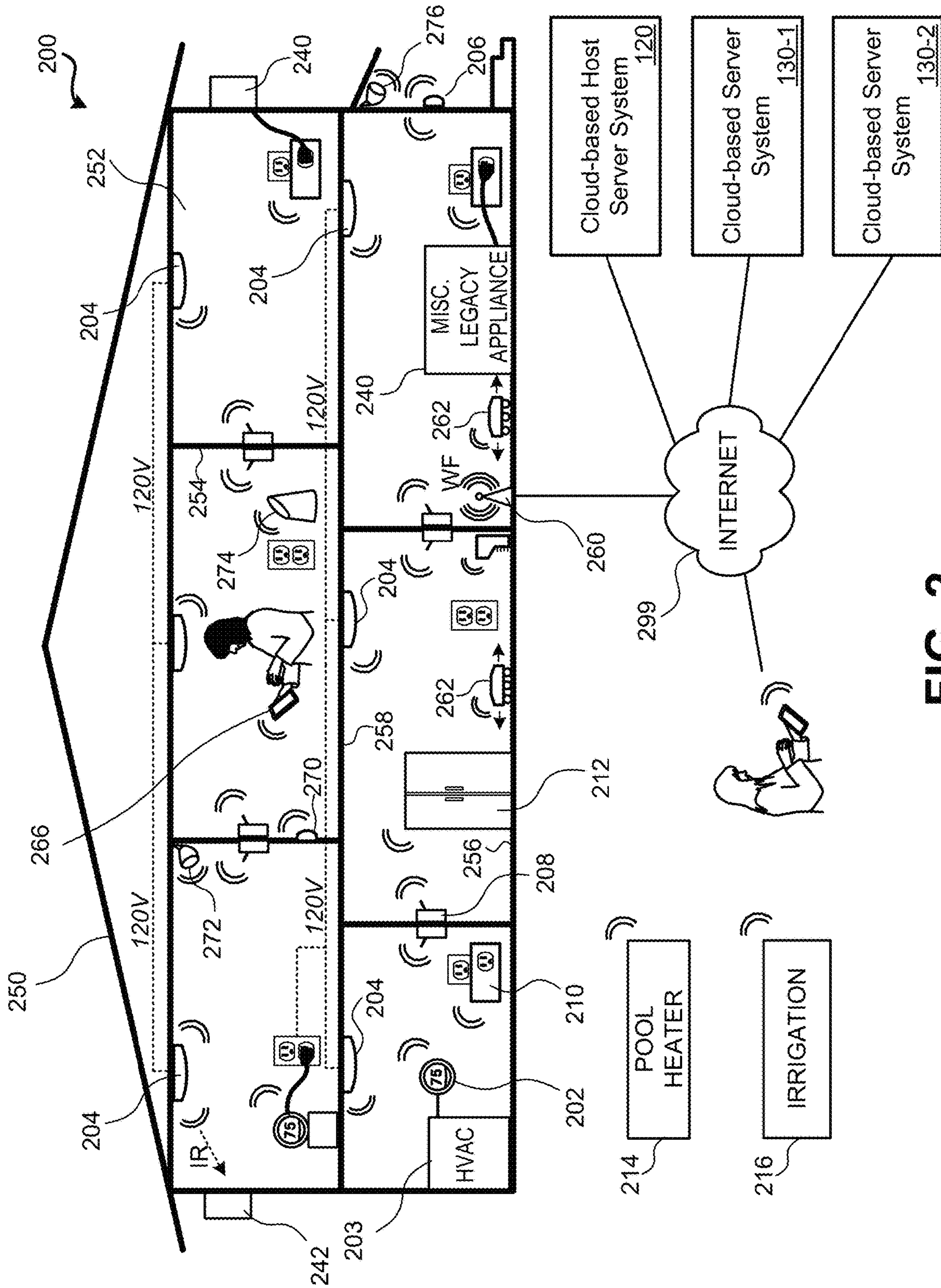

200
250
252
254
256
258
120V
120V
120V
120V
204
204
204
204
204
204
240
240
242
276
206
262
262
WF
260
274
266
270
272
212
208
210
202
203
IR
75
75
MISC.
LEGACY
APPLIANCE
HVAC
Cloud-based Host
Server System
120
Cloud-based Server
System
130-1
Cloud-based Server
System
130-2
INTERNET
299
POOL
HEATER
214
IRRIGATION
216

FIG. 2

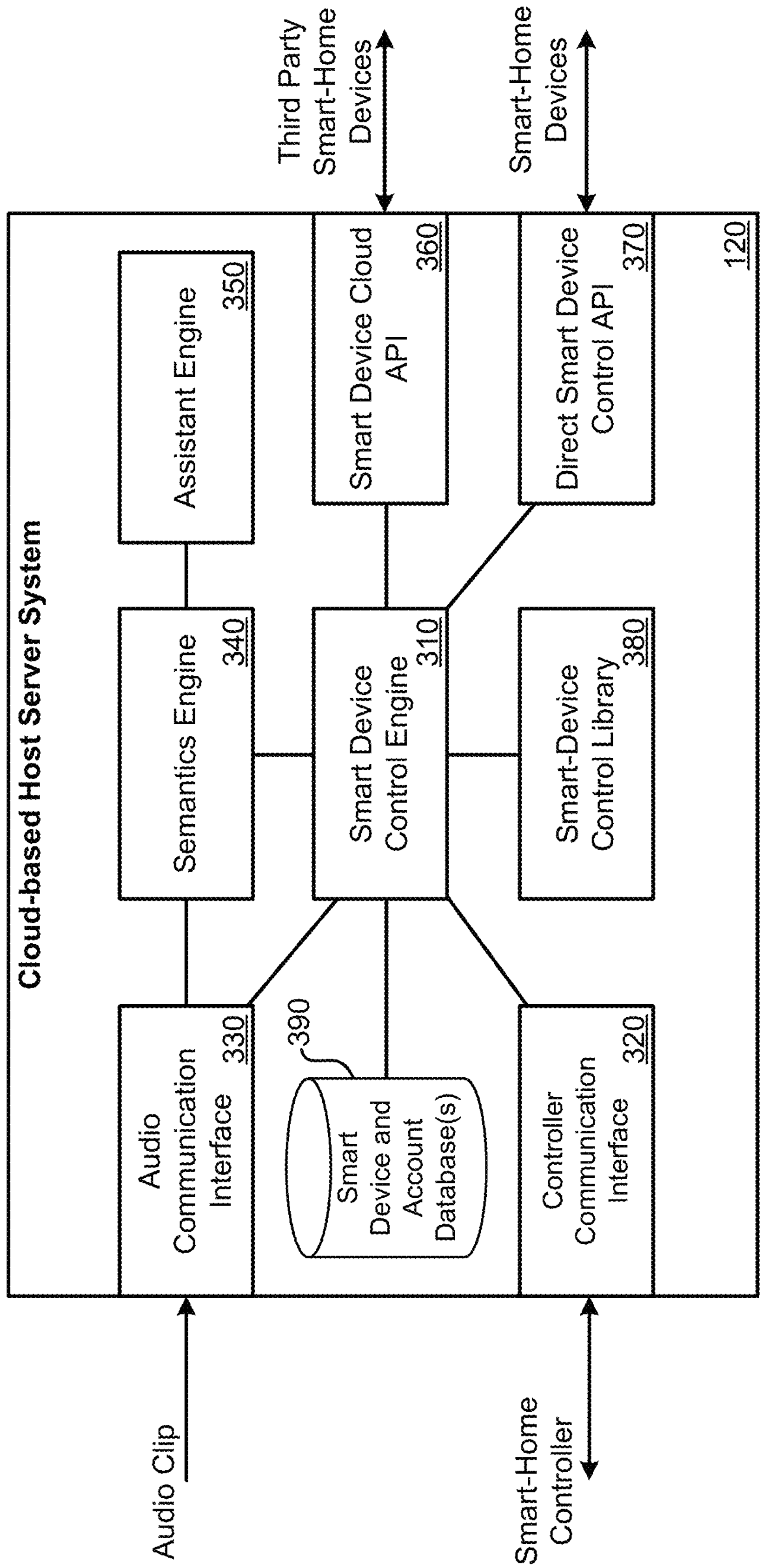

Cloud-based Host Server System
120
Audio Clip
Smart-Home Controller
Third Party Smart-Home Devices
Smart-Home Devices
Audio Communication Interface 330
Smart Device and Account Database(s)
390
Controller Communication Interface 320
Semantics Engine 340
Smart Device Control Engine 310
Smart-Device Control Library 380
Assistant Engine 350
Smart Device Cloud API 360
Direct Smart Device Control API 370

FIG. 3

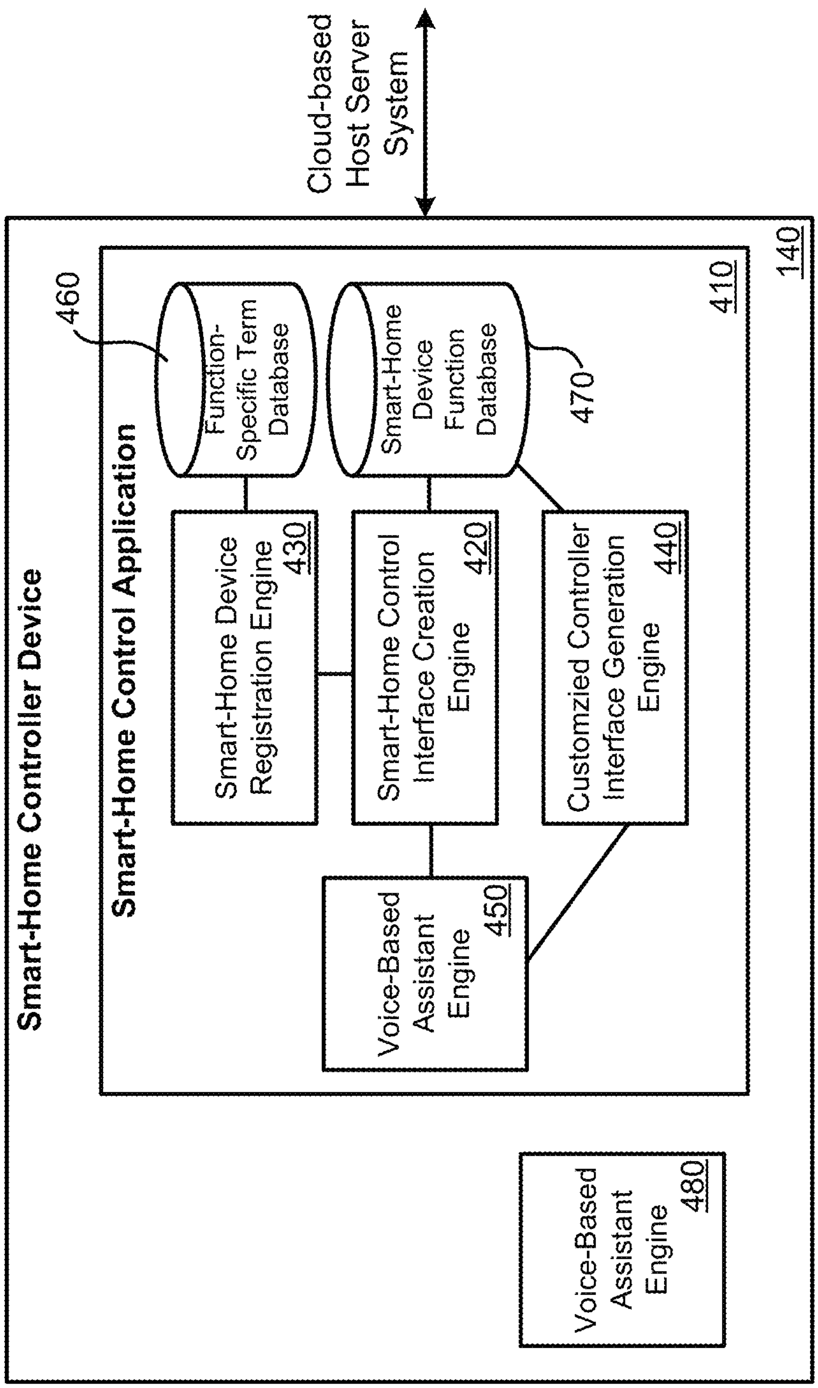

Cloud-based Host Server System
Smart-Home Controller Device
140
Smart-Home Control Application
410
460
Function-Specific Term Database
Smart-Home Device Function Database
470
Smart-Home Device Registration Engine 430
Smart-Home Control Interface Creation Engine 420
Customzied Controller Interface Generation Engine 440
Voice-Based Assistant Engine 450
Voice-Based Assistant Engine 480

FIG. 4

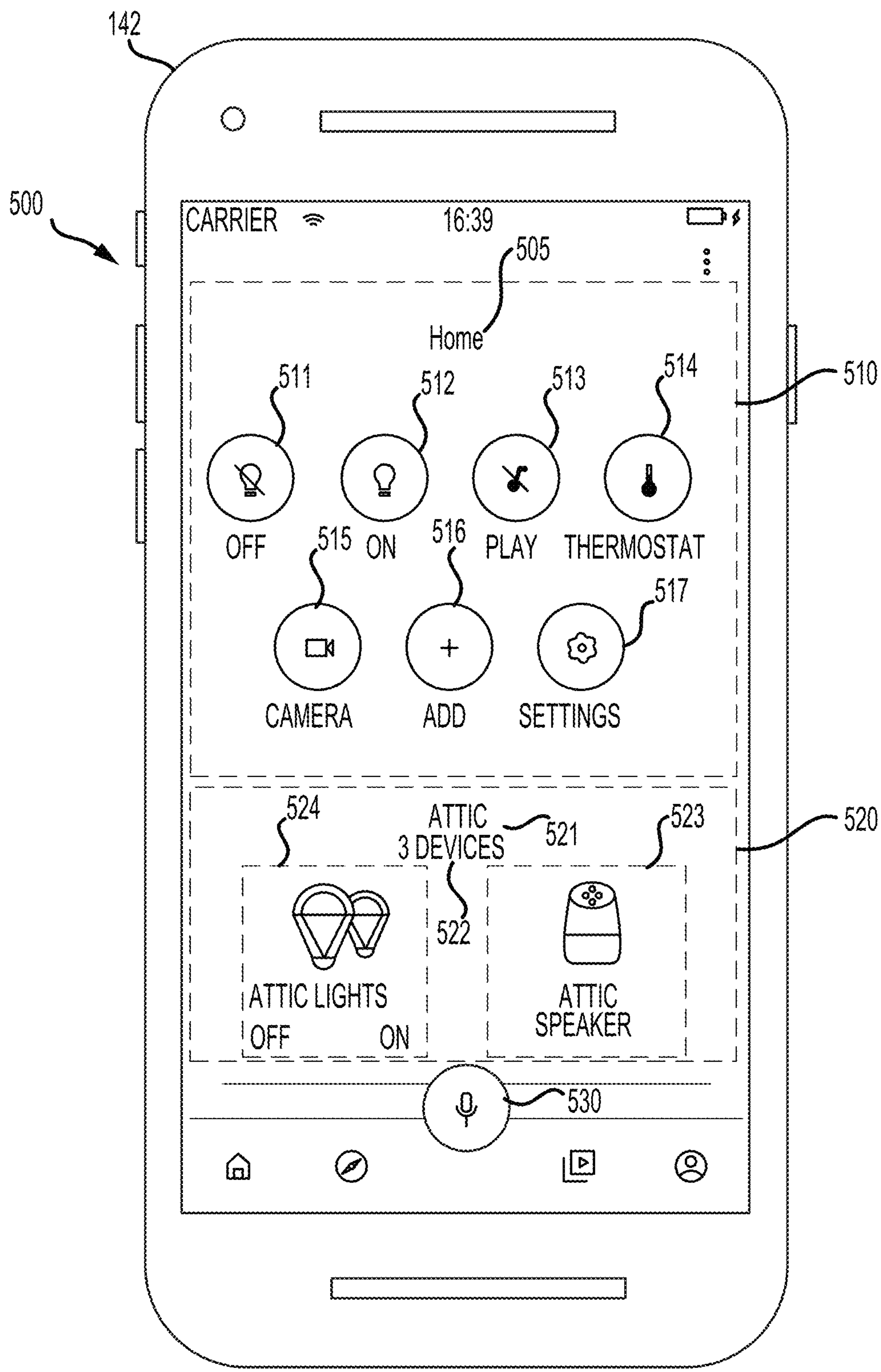

142
500
CARRIER
16:39
505
Home
510
511
512
513
514
OFF
ON
PLAY
THERMOSTAT
515
516
517
CAMERA
ADD
SETTINGS
520
524
ATTIC
3 DEVICES
521
523
522
ATTIC LIGHTS
OFF
ON
ATTIC
SPEAKER
530

FIG. 5

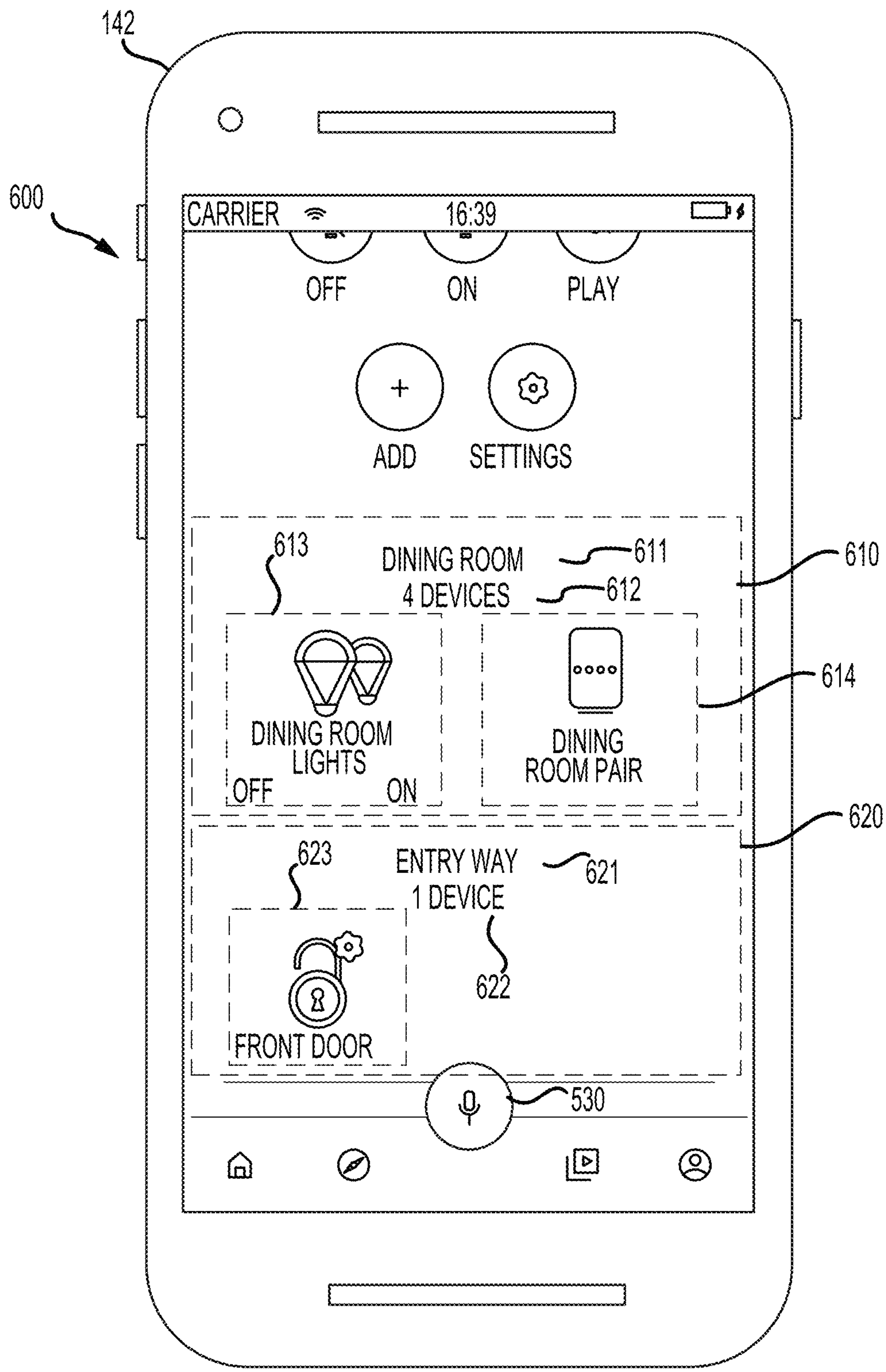

142
600
CARRIER
16:39
OFF
ON
PLAY
+
ADD
SETTINGS
613
DINING ROOM
611
4 DEVICES
612
610
DINING ROOM LIGHTS
OFF
ON
DINING ROOM PAIR
614
620
623
ENTRY WAY
621
1 DEVICE
622
FRONT DOOR
530

FIG. 6

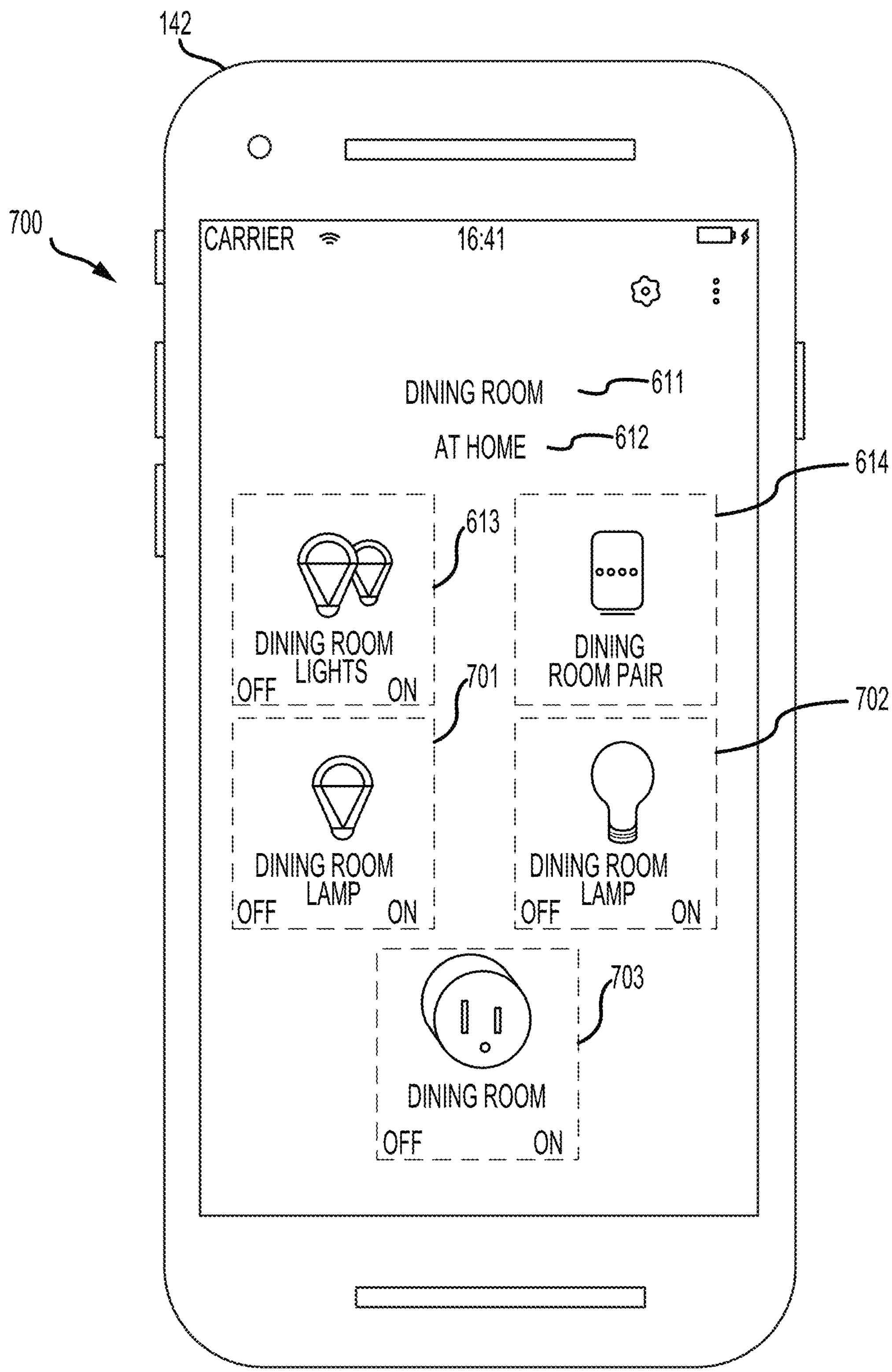

142
700
CARRIER
16:41
DINING ROOM
611
AT HOME
612
614
613
DINING ROOM LIGHTS
OFF
ON
DINING ROOM PAIR
701
702
DINING ROOM LAMP
OFF
ON
DINING ROOM LAMP
OFF
ON
703
DINING ROOM
OFF
ON

FIG.7

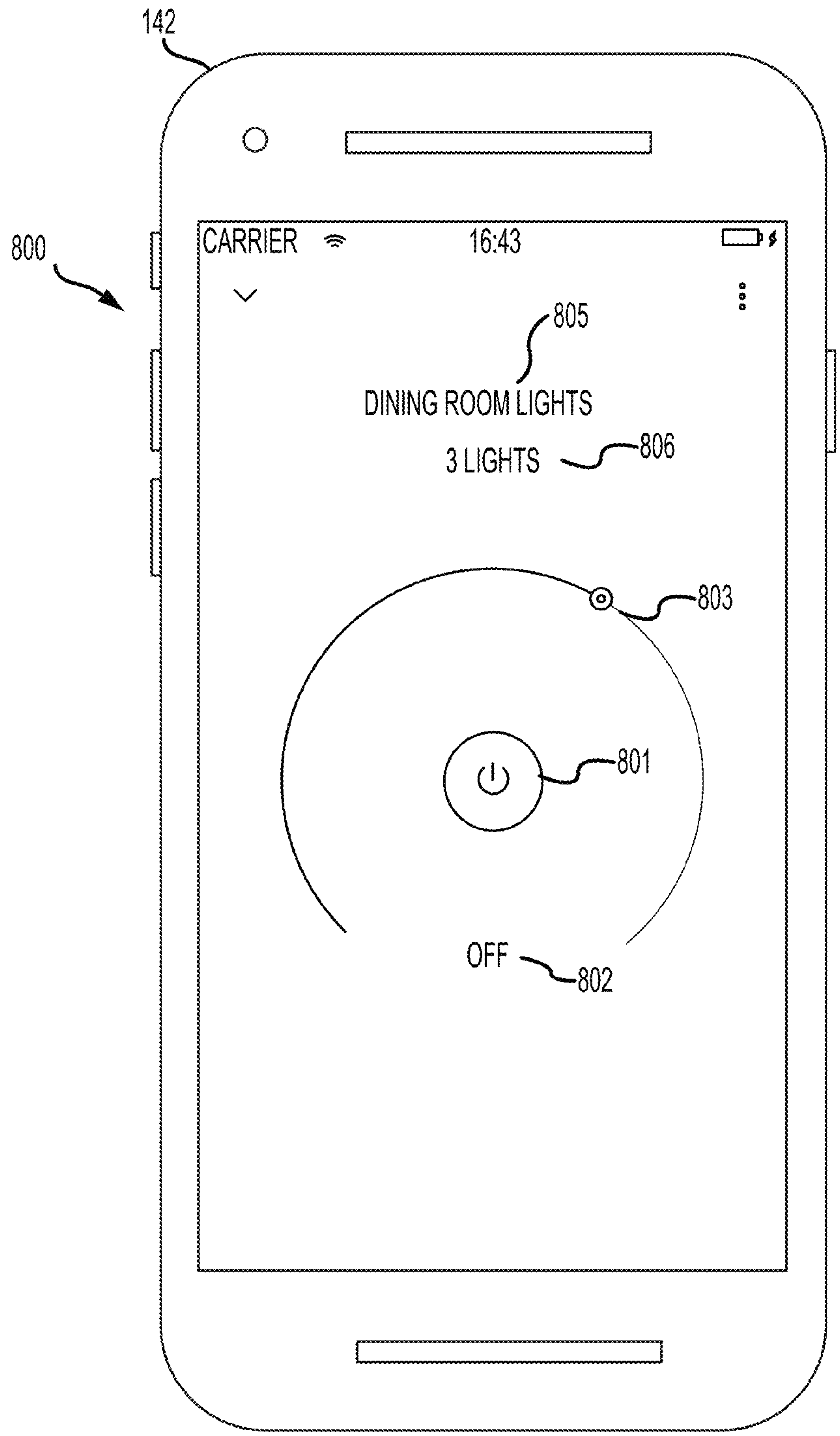

142
800
CARRIER
16:43
805
DINING ROOM LIGHTS
3 LIGHTS
806
803
801
OFF
802

FIG. 8

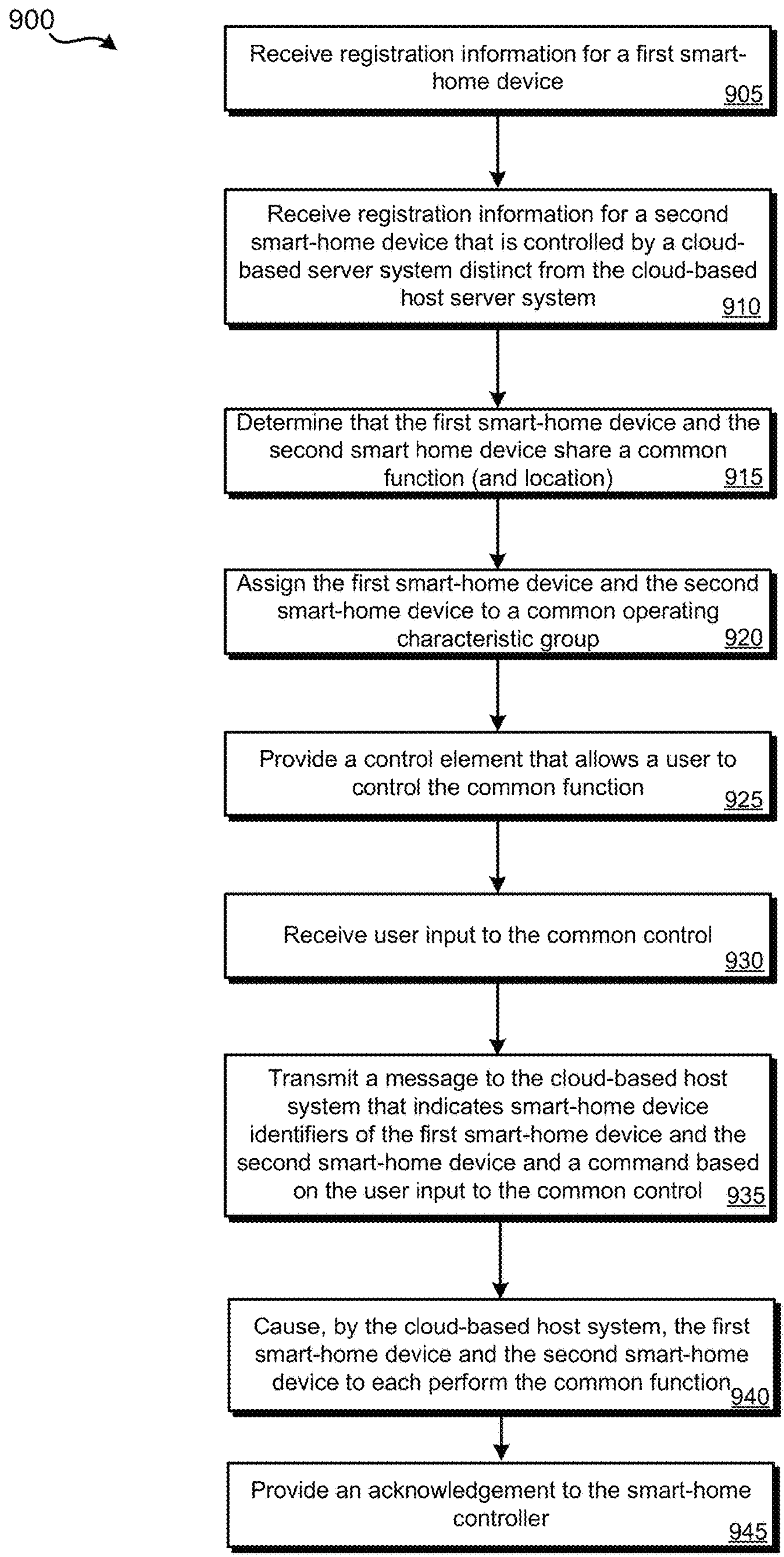

900
Receive registration information for a first smart-home device 905
Receive registration information for a second smart-home device that is controlled by a cloud-based server system distinct from the cloud-based host server system 910
Determine that the first smart-home device and the second smart home device share a common function (and location) 915
Assign the first smart-home device and the second smart-home device to a common operating characteristic group 920
Provide a control element that allows a user to control the common function 925
Receive user input to the common control 930
Transmit a message to the cloud-based host system that indicates smart-home device identifiers of the first smart-home device and the second smart-home device and a command based on the user input to the common control 935
Cause, by the cloud-based host system, the first smart-home device and the second smart-home device to each perform the common function 940
Provide an acknowledgement to the smart-home controller 945

FIG. 9

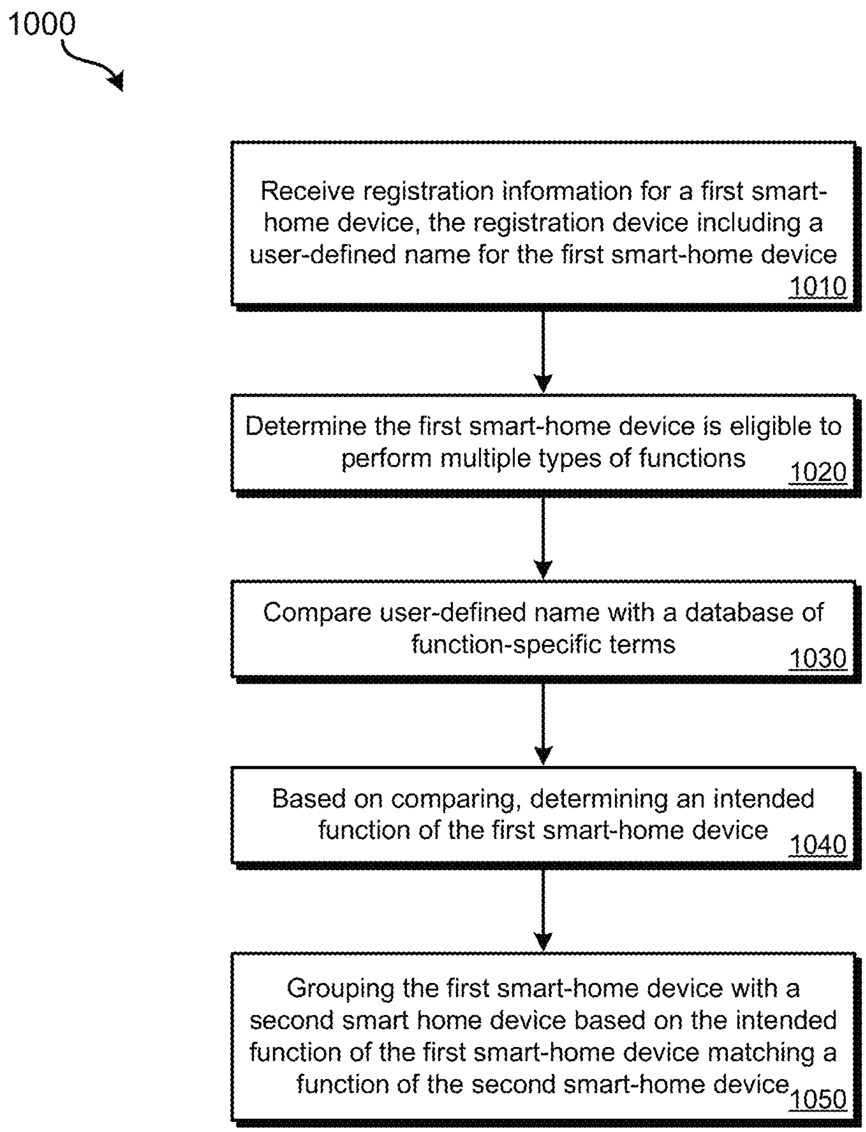

1000
Receive registration infomation for a first smart-home device, the registration device including a user-defined name for the first smart-home device
1010
Determine the first smart-home device is eligible to perform multiple types of functions
1020
Compare user-defined name with a database of function-specific terms
1030
Based on comparing, determining an intended function of the first smart-home device
1040
Grouping the first smart-home device with a second smart home device based on the intended function of the first smart-home device matching a function of the second smart-home device
1050

FIG. 10

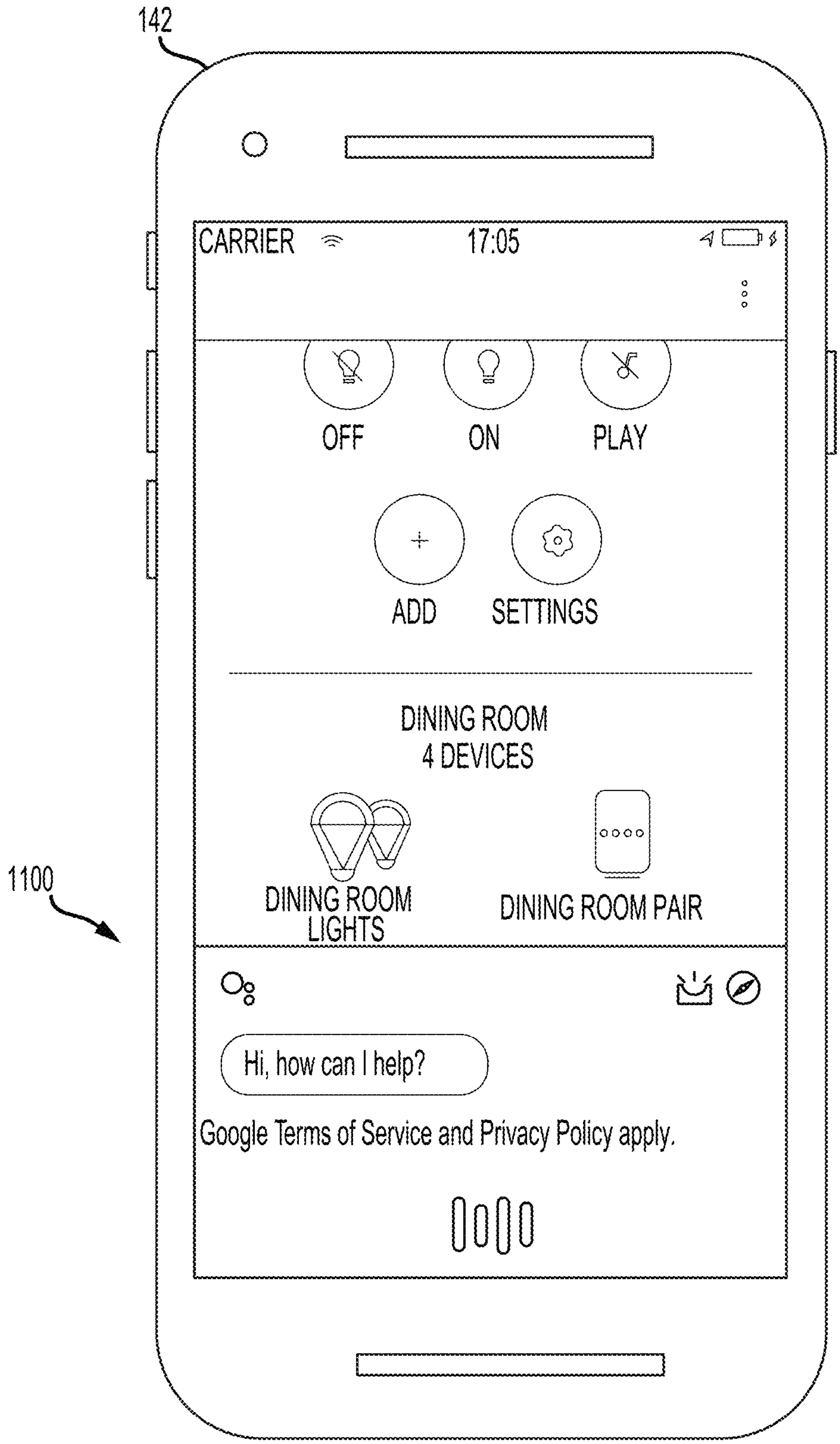

142
CARRIER
17:05
OFF
ON
PLAY
ADD
SETTINGS
DINING ROOM
4 DEVICES
DINING ROOM LIGHTS
DINING ROOM PAIR
Hi, how can I help?
Google Terms of Service and Privacy Policy apply.
1100

FIG. 11

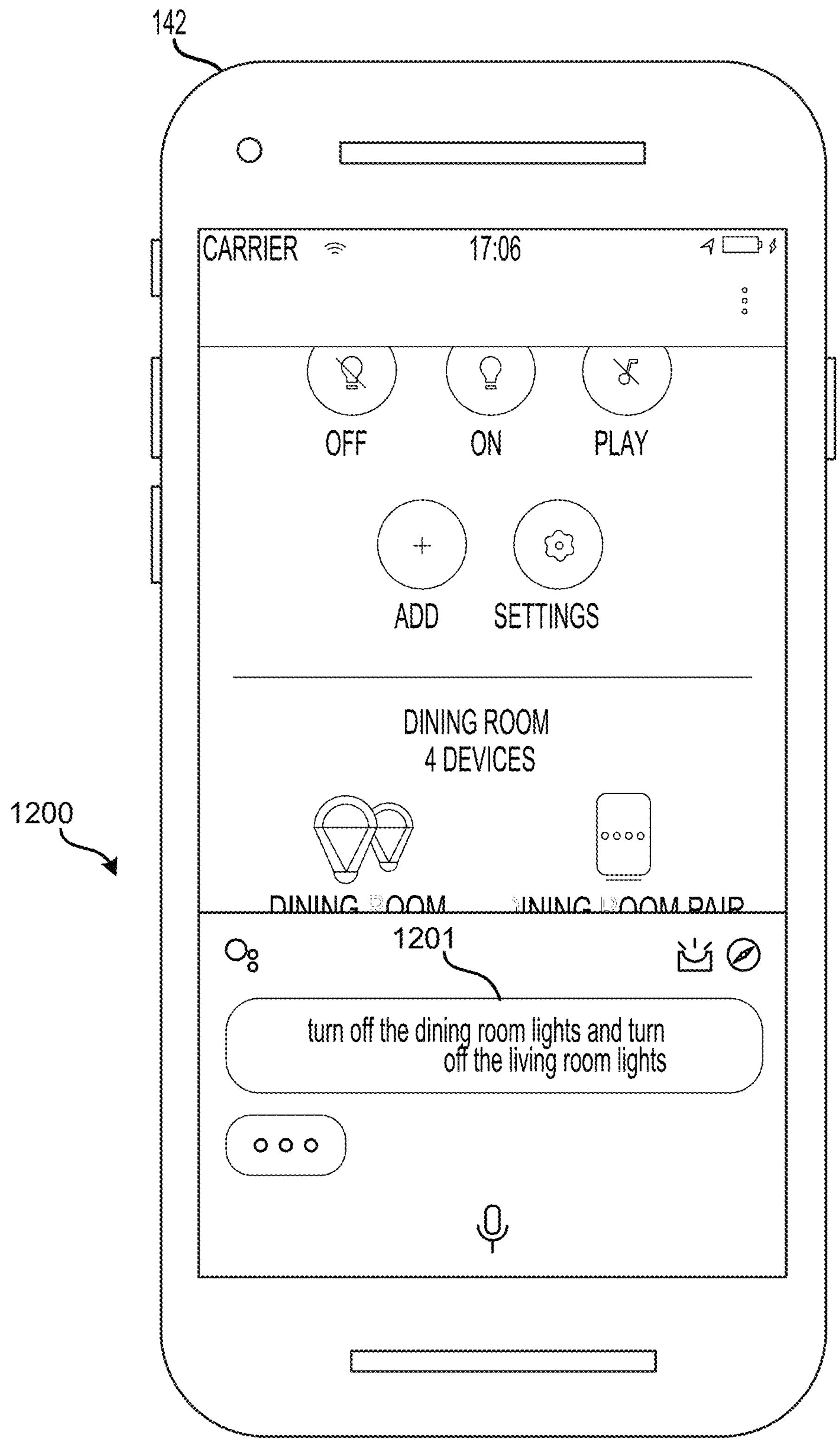

142
CARRIER
17:06
OFF
ON
PLAY
ADD
SETTINGS
DINING ROOM
4 DEVICES
1200
1201
turn off the dining room lights and turn off the living room lights

FIG. 12

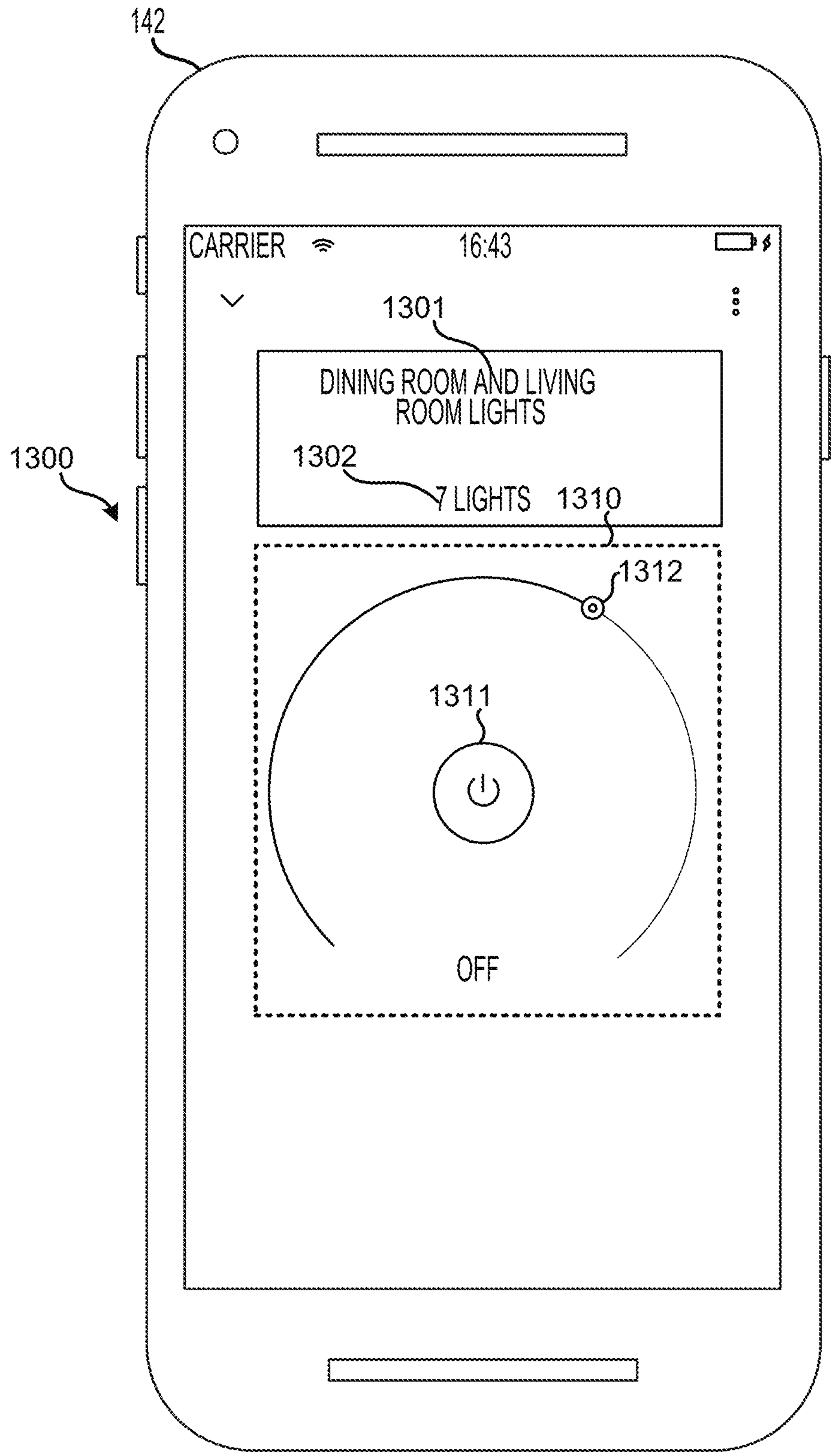

142
CARRIER
16:43
1301
DINING ROOM AND LIVING ROOM LIGHTS
1302
7 LIGHTS
1310
1300
1312
1311
OFF

FIG. 13

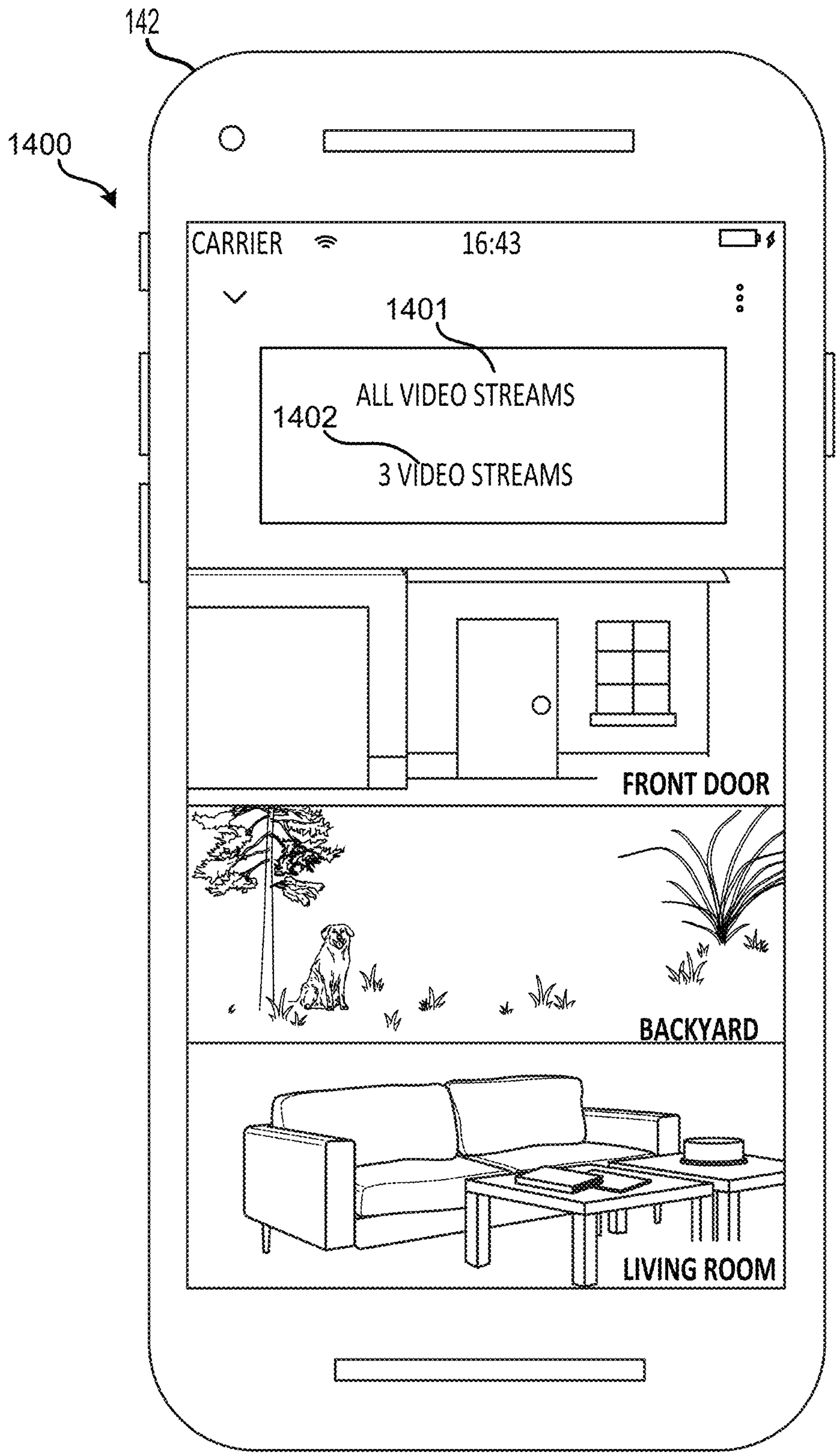

142
1400
CARRIER
16:43
1401
ALL VIDEO STREAMS
1402
3 VIDEO STREAMS
FRONT DOOR
BACKYARD
LIVING ROOM

FIG. 14

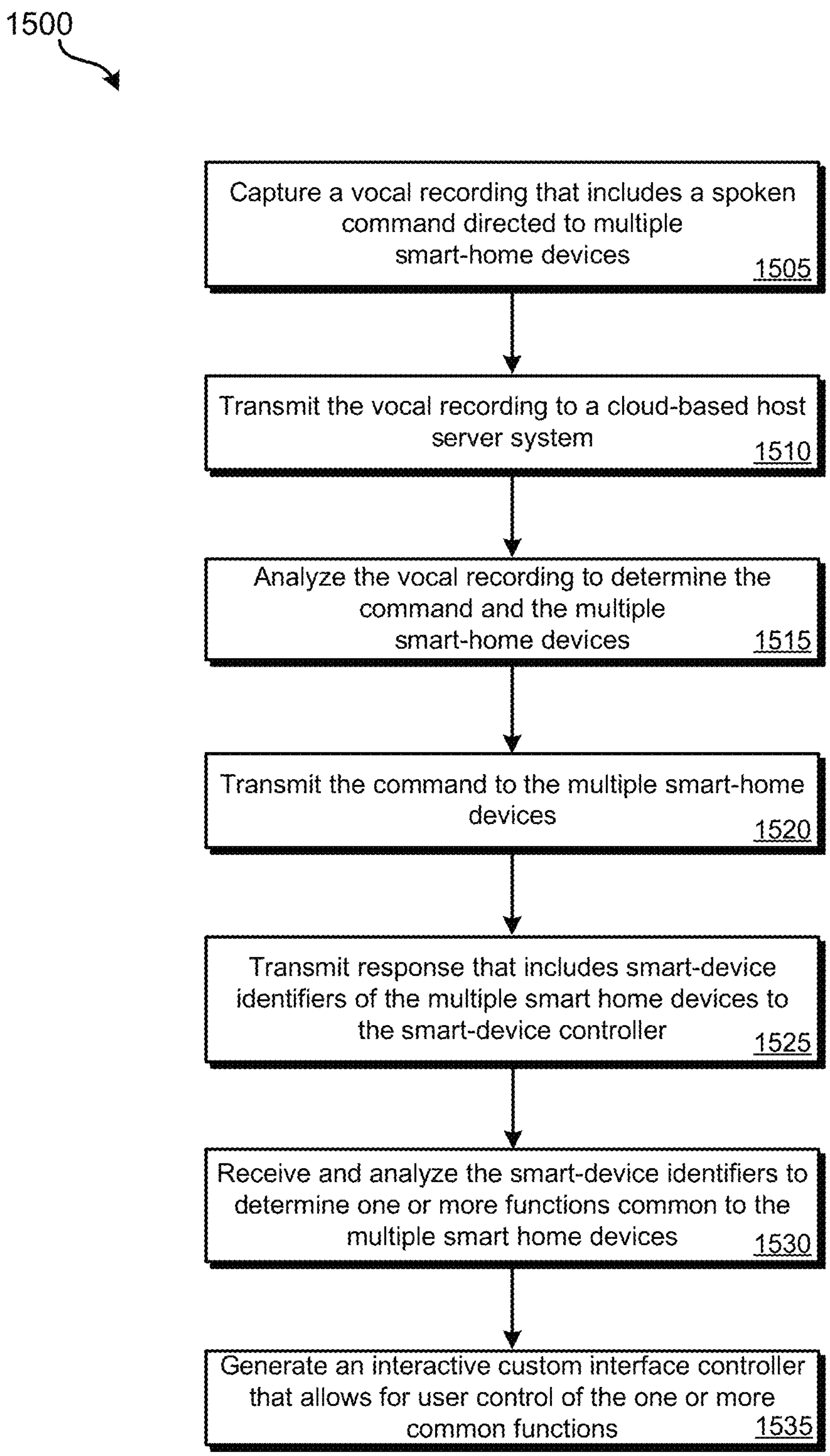

1500
Capture a vocal recording that includes a spoken command directed to multiple smart-home devices
1505
Transmit the vocal recording to a cloud-based host server system
1510
Analyze the vocal recording to determine the command and the multiple smart-home devices
1515
Transmit the command to the multiple smart-home devices
1520
Transmit response that includes smart-device identifiers of the multiple smart home devices to the smart-device controller
1525
Receive and analyze the smart-device identifiers to determine one or more functions common to the multiple smart home devices
1530
Generate an interactive custom interface controller that allows for user control of the one or more common functions
1535

FIG. 15

MULTI-SOURCE SMART-HOME DEVICE CONTROL

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/709,077, filed Mar. 30, 2022, entitled "Customized Interface based on Vocal Input," which is a continuation of U.S. patent application Ser. No. 16/154,466, filed Oct. 8, 2018, entitled "Multi-Source Smart-Home Device Control," the entire disclosures of which are hereby incorporated by reference for all purposes. This application is related U.S. patent application Ser. No. 16/154,512, filed on Oct. 8, 2018, entitled "Customized Interface based on Vocal Input," the entire disclosure of which is hereby incorporated by reference for all purposes.

BACKGROUND

Smart-home devices, such as smart outlets, smart lights, smart speakers, home assistant devices, smart thermostats, and smart hazard detectors are becoming common in homes worldwide. Effectively allowing a single controller device to control various smart-home devices that have varying capabilities and that receive commands from different cloud-based server systems can result in significant technical challenges.

SUMMARY

Various embodiments are described related to method for integrating control of multiple cloud-based smart-home devices. In some embodiments, a method for integrating control of multiple cloud-based smart-home devices is described. The method may include receiving, by an application being executed by a mobile device, a first set of smart-home device registration information for a first smart-home device. At least one function of the first smart-home device may be controlled via communication with a first cloud-based server system. The method may include receiving, by the application being executed by the mobile device, a second set of smart-home device registration information for a second smart-home device. At least one function of the second smart-home device may be controlled via communication with a second cloud-based server system distinct from the first cloud-based server system. The method may include determining, by the application being executed by the mobile device, that the first smart-home device and the second smart-home device share a common function. The method may include assigning the first smart-home device controlled via communication with the first cloud-based server system and the second smart-home device controlled via communication with the second cloud-based server system to a common operating characteristic group based on the common function being shared by the first smart-home device and the second smart-home device. The method may include providing, by the application being executed by the mobile device, a control element that allows for control of smart-home devices with the common operating characteristic group. The control element controls the common function at the first smart-home device via the first cloud-based server system and at the second smart-home device via the second cloud-based server system.

Embodiments of such a method may include one or more of the following features: providing separate controls for the first smart-home device and the second smart-home device.

The method may include providing a second control element concurrently with the control element that controls a second function that may only be performed at one of the first smart-home device and the second smart-home device. Receiving, by the application being executed by the mobile device, the first set of smart-home device registration information for the first smart-home device may include receiving, by the application, from a user, a user-defined name. Determining that the first smart-home device and the second smart-home device share the common function may include comparing the user-defined name to database of function-specific terms and determining the function of the first smart-home device based on comparing the user-defined name to the database of function-specific terms. Determining that the first smart-home device and the second smart-home device share the common function may include determining the function of the first smart-home device that was determined based on the user-defined name matches the function of the second smart-home device. The first smart-home device may be a smart outlet plug. Receiving the first set of smart-home device registration information for the first smart-home device may include receiving the user-defined name that includes a term meaning "light." The first smart-home device that may be the smart outlet plug may be assigned to a lighting group. The first set of smart-home device registration information may include an indication of a location and the second set of smart-home device registration information may include the indication of the location. Assigning the first smart-home device controlled via communication with the first cloud-based server system and the second smart-home device controlled via communication with the first cloud-based server system to the common operating characteristic group may be further based on the first smart-home device and the second smart-home device being mapped to the location.

In some embodiments, a system for integrating control of multiple cloud-based smart-home devices is described. The system may include an application executed by a mobile device. The application may cause one or more processors of the mobile device to receive a first set of smart-home device registration information for a first smart-home device. At least one function of the first smart-home device may be controlled via communication with a first cloud-based server system. The application may cause one or more processors of the mobile device to receive a second set of smart-home device registration information for a second smart-home device. At least one function of the second smart-home device may be controlled via communication with a second cloud-based server system distinct from the first cloud-based server system. The application may cause one or more processors of the mobile device to determine that the first smart-home device and the second smart-home device share a common function. The application may cause one or more processors of the mobile device to assign the first smart-home device controlled via communication with the first cloud-based server system and the second smart-home device controlled via communication with the second cloud-based server system to a common operating characteristic group based on the common function being shared by the first smart-home device and the second smart-home device. The application may cause one or more processors of the mobile device to output a control element that allows for control of smart-home devices with the common operating characteristic group. The control element may control the common function at the first smart-home device via the first cloud-based server system and at the second smart-home device via the second cloud-based server system.

Embodiments of such a system may include one or more of the following features: the application may further cause the one or more processors of the mobile device to provide separate controls for the first smart-home device and the second smart-home device. The application may further cause the one or more processors of the mobile device to provide a second control element concurrently with the control element that controls a second function that may only be performed at one of the first smart-home device and the second smart-home device. The application causing the one or more processors to receive the first set of smart-home device registration information for the first smart-home device may include the application causing the one or more processors of the mobile device to: receive, from a user, a user-defined name. The application causing the one or more processors to determine that the first smart-home device and the second smart-home device share the common function may include the application causing the one or more processors of the mobile device to compare the user-defined name to database of function-specific terms and determine the function of the first smart-home device based on comparing the user-defined name to the database of function-specific terms. The application causing the one or more processors to determine that the first smart-home device and the second smart-home device share the common function may include the application causing the one or more processors of the mobile device to determine the function of the first smart-home device that was determined based on the user-defined name matches the function of the second smart-home device. The first smart-home device may be a smart outlet plug. The application causing the one or more processors to receive the first set of smart-home device registration information for the first smart-home device may include the application causing the one or more processors of the mobile device to receive the user-defined name that includes a term meaning "light." The first smart-home device that may be the smart outlet plug may be assigned to a lighting group. The first set of smart-home device registration information may include an indication of a location and the second set of smart-home device registration information may include the indication of the location.

In some embodiments, a system for integrating control of multiple cloud-based smart-home devices is described. The system may include a cloud-based host server system. The system may include an application executed by a mobile device. One or more processors of the cloud-based host server system and the mobile device may be configured to receive a first set of smart-home device registration information for a first smart-home device. At least one function of the first smart-home device may be controlled via communication with a first cloud-based server system. One or more processors of the cloud-based host server system and the mobile device may be configured to receive a second set of smart-home device registration information for a second smart-home device. At least one function of the second smart-home device may be controlled via communication with a second cloud-based server system distinct from the first cloud-based server system. One or more processors of the cloud-based host server system and the mobile device may be configured to determine that the first smart-home device and the second smart-home device share a common function. One or more processors of the cloud-based host server system and the mobile device may be configured to assign the first smart-home device controlled via communication with the first cloud-based server system and the second smart-home device controlled via communication with the second cloud-based server system to a common operating characteristic group based on the common function being shared by the first smart-home device and the second smart-home device. One or more processors of the cloud-based host server system and the mobile device may be configured to output a control element by the mobile device that allows for control of smart-home devices with the common operating characteristic group. The control element may control the common function at the first smart-home device via the first cloud-based server system and at the second smart-home device via the second cloud-based server system.

BRIEF DESCRIPTION OF THE DRAWINGS

A further understanding of the nature and advantages of various embodiments may be realized by reference to the following figures. In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

FIG. **1** illustrates a block diagram of an embodiment of a smart-home system.

FIG. **2** illustrates an embodiment of a smart-home with additional detail regarding smart-home devices.

FIG. **3** illustrates a block diagram of an embodiment of a cloud-based host server system.

FIG. **4** illustrates a block diagram of an embodiment of a smart-home controller device.

FIG. **5** illustrates an embodiment of an interface that may be presented by a smart-home controller device.

FIG. **6** illustrates an embodiment of an interface that may be presented by a smart-home controller device that includes grouped smart-home devices based on location.

FIG. **7** illustrates an embodiment of an interface that may be presented by a smart-home controller device for a particular group.

FIG. **8** illustrates an embodiment of a control that may be presented that can be used to control multiple smart-home devices.

FIG. **9** illustrates an embodiment of a method for integrating control of multiple cloud-based smart-home devices.

FIG. **10** illustrates an embodiment of a method for determining how to group a smart-home device.

FIG. **11** illustrates an embodiment of a control interface that may be presented when ready to capture an audio sample.

FIG. **12** illustrates an embodiment of a control interface that may be presented while capturing an audio sample.

FIG. **13** illustrates an embodiment of a control interface that can control a common function across smart-home devices indicated in the audio sample.

FIG. **14** illustrates an embodiment of an interface that allows a user to view video streams indicated in the audio sample.

FIG. **15** illustrates an embodiment of a method for using captured voice to generate an interactive custom interface controller.

DETAILED DESCRIPTION

Smart-home devices may be controlled using a smart-home controller device. A smart-home controller device may be a wireless or mobile computerized device, such as a smartphone, tablet computer, laptop computer, dedicated home-controller device, or some other form of computerized device, that may execute an application that allows a user to control the functionality of one or more smart-home devices. The smart-home controller device may communicate with a cloud-based host server system. The cloud-based host system may communicate with various smart-home devices and may communicate with other cloud-based system to control smart-home devices that cannot be directly controlled by the cloud-based host system (e.g., smart devices manufactured and sold by other manufacturers).

The smart-home controller device may provide an interface that allows for smart-home devices to be controlled together and/or using a single input regardless of whether each smart-home device is controlled by the cloud-based host system or through another cloud-based system. The smart-home controller device may allow for smart-devices to be grouped based on location and/or function. Within such an arrangement, a common control element may be provided that allows a user to provide a single input that indicates a command that is to be implemented at multiple smart-home devices, regardless of whether the smart-home devices are controlled directly by the cloud-based host server system and/or through a separate cloud-based server system.

The smart-home controller device may additionally be configured to reclassify various types of smart-home devices based on user behavior. Such a reclassification may allow the smart-home controller device to determine when to send a command to a smart-home device more accurately and effectively. When a smart-home device is initially registered with the smart-home controller device or the registration is edited, a user may name the smart-home device and/or identify characteristics of the smart-home device. The words (and/or parts of words) submitted as part of the name may be compared to a smart-home device function database. The use of a particular word, part of a word, or phrase within a name of a smart-home device may be indicative of the smart-home device being used to perform a particular function. As an example, a smart outlet plug, if given the name of "dining room light" by a user, the smart outlet plug can be expected to control whether power is provided to a light located within the dining room. The smart-home controller device may classify the smart outlet plug in a lighting group (e.g., a dining room lighting group) because while the smart device is a smart outlet plug, it is effectively functioning as a controller for a light. Thus, if a user provides a command that indicates all of the lights within the dining room should be turned on or off, the smart-home controller device or cloud-based host server system may send the command to the smart outlet plug since it controls a light.

Additionally or alternatively, voice-based commands may be provided by a user to the smart-home controller device to control the functionality of multiple smart-home devices. The user may speak a command that does not correspond to a particular smart-home device or a predefined group of smart-home devices. For example, a user may say "turn on the dining room lights and the kitchen fan" or "turn off the dining room and the living room lights." Embodiments detailed herein detail how a custom controller may be generated that controls at least one common function across all of the smart-home devices indicated in the spoken command. This interactive custom controller may allow a user to further change the state of the multiple smart-home devices that were the subject of the initial spoken command. The custom controller may include a graphical representation of a power button that, referring to the first example, controls both the dining room lights and the kitchen fan or, referring to the second example, all of the smart lights in the dining room and the living room. This custom controller may present some or all of the functions that all of the smart-home devices that were indicated in the spoken command have in common. Additionally or alternatively, the custom controller may present one or more functions that can only be performed at some of the smart-home devices indicated in the spoken command. For example, referring back to the example of the dining room lights and the kitchen fan, the custom controller may include a dimmer control that does not affect the state of the fan but allows at least a subset of the lights in the dining room to be dimmed via additional user input.

Further detail regarding the above embodiments and additional embodiments is provided in relation to the figures. While the following description is focused on smart-home devices, it should be understood that other forms of smart-devices may be controlled as detailed other than smart-home devices. For instance, smart devices in a factory, office, deployed in the field, or some other form of smart devices may be controlled as detailed herein. FIG. **1** illustrates a block diagram of an embodiment of a smart-home system **100**. Smart-home system **100** may include smart-home devices **110** (**110-1**, **110-2**, **110-3**, **110-4**, **110-5**); cloud-based host server system **120**; cloud-based server systems **130** (**130-1**, **130-2**); and smart-home controller devices **140**.

Smart-home devices **110** represent various smart-home devices that may be present in a particular structure **105** (e.g., a home, an apartment, a condominium, an office, a warehouse, a factory, etc.). Smart-home devices **110** may have differing sensors on-board and may or may not have the ability to output information (e.g., via sound, light, or an electronic display). Specific forms of smart-home devices **110** are detailed in relation to FIG. **2**. In the illustrated embodiment of system **100**, five smart-home devices **110** are presented. This number of smart-home devices is for example purposes only; in other embodiments, fewer or greater numbers of smart-home devices may be present within or at structure **105**. Each of smart-home devices **110** may be able to perform some form of function in response to a command received from an associated form of cloud-based server system (**120**, **130**). For example, possible commands may be instructions to turn on, turn off, dim, lock, unlock, play, pause, stop, turn the volume up, turn the volume down, mute, hush, turn the temperature up, turn the temperature down, etc.

Smart-home devices **110** may have been acquired from varying sources. For example, a first entity (e.g., manufacturer) may produce smart-home devices **110-1** and **110-2**, a second entity may produce smart-home device **110-3**, and a third entity may produce smart-home devices **110-4** and **110-5**. Such an arrangement in which smart-home devices from multiple entities is present within structure **105** may be fairly common: while a user may desire the feature set of one type of smart-home device, for another type of smart-home device, another entity may provide benefits that the user prefers.

Each smart-home device of smart-home devices **110** may communicate with a form of a cloud-based server system (**120**, **130**). That is, each entity that provides smart-home devices may operate or have operated on its behalf a cloud-based server system that can receive data from its associated smart-home devices and send data (e.g., commands) to its associated smart-home devices. Cloud-based host server system **120** may communicate directly with smart-home devices **110-1** and **110-2**. In this description, "directly" refers to the a server system communicating with a smart-home device via one or more networks without any additional intervening smart-home server systems. For example, the communication between smart-home device **110-1** and cloud-based host server system **120** may occur via a local wireless network (WLAN) provided by a router within structure **105**. The router may be connected to an Internet service provider (ISP) that provides a wired or wireless communication link to the Internet. Cloud-based host server system **120** may be connected with the Internet, thus allowing data to be passed between smart-home device **110-1** and cloud-based host server system **120**.

For other entities, separate cloud-based server systems may control associated smart-home devices: smart-home device **110-3** communicates directly with cloud-based server system **130-1** and smart-home devices **110-4** and **110-5** communicate directly with cloud-based server system **130-2**. Cloud-based host server system **120** can additionally communicate with other cloud-based server systems **130** through one or more application programming interfaces (APIs). Therefore, cloud-based host server system **120** can receive data from and control smart-home devices **110-3** through **110-5**, albeit not directly because communication via cloud-based server systems **130-1** and **130-2** is used. Further detail regarding cloud-based host server system **120** is provided in relation to FIG. **3**.

Smart-home controller devices **140** may serve as an interface for one or more users to interact with cloud-based host server system **120** to receive data from and control smart-home devices **110**. Some or all of smart-home controller devices **140** may each receive auditory commands in the form of speech spoken by a user. Further, each of smart-home controller devices **140** may have a display or touchscreen that allows graphical user interfaces (GUIs) to be presented to the users. Smart-home controller devices **140** can include: smartphone **142**; tablet computer **144**; and laptop computer **146**. Other forms of smart-home controller devices **140** may be possible, such as a dedicated smart-home device controller computerized device, a desktop computer, or a gaming device. Further detail regarding a smart-home controller device is provided in relation to FIG. **4**.

FIG. **2** illustrates an embodiment of a smart-home environment **200** with additional detail regarding smart-home devices. Smart-home **200** represents an embodiment of structure **105** that includes multiple smart-home devices. The depicted smart-home environment **200** includes a structure **250**, which can include, e.g., a house, office building, garage, or mobile home. It will be appreciated that devices can also be integrated into a smart-home environment **200** that does not include an entire structure **250**, such as an apartment, condominium, or office space. Further, the smart-home environment can control and/or be coupled to devices outside of the actual structure **250**. Indeed, several devices in the smart-home environment need not physically be within the structure **250** at all. For example, a device controlling a pool heater **214** or irrigation monitor **216** can be located outside of the structure **250**.

The depicted structure **250** includes a plurality of rooms **252**, separated at least partly from each other via walls **254**. The walls **254** can include interior walls or exterior walls. Each room can further include a floor **256** and a ceiling **258**. Devices can be mounted on, integrated with and/or supported by a wall **254**, floor **256** or ceiling **258**.

In some embodiments, the smart-home environment **200** of FIG. **2** includes a plurality of smart-home devices, including intelligent, multi-sensing, network-connected devices, that can integrate seamlessly with each other and/or with a central server or a cloud-computing system to provide any of a variety of useful smart-home objectives. The smart-home environment **200** may include one or more intelligent, multi-sensing, network-connected thermostats **202** (hereinafter referred to as "smart thermostats **202**"), one or more intelligent, network-connected, hazard detectors **204**, and one or more intelligent, multi-sensing, network-connected entryway interface devices **206** (hereinafter referred to as "smart doorbells **206**"). According to embodiments, the smart thermostat **202** detects ambient climate characteristics (e.g., temperature and/or humidity) and controls a HVAC system **203** accordingly. The hazard detector **204** may detect the presence of a hazardous substance or a substance indicative of a hazardous substance (e.g., smoke, fire, or carbon monoxide). The smart doorbell **206** may detect a person's approach to or departure from a location (e.g., an outer door), control doorbell functionality, announce a person's approach or departure via audio or visual means, or control settings on a security system (e.g., to activate or deactivate the security system when occupants go and come).

In some embodiments, the smart-home environment **200** of FIG. **2** further includes one or more intelligent, multi-sensing, network-connected wall switches **208** (hereinafter referred to as "smart wall switches **208**"), and/or with one or more intelligent, multi-sensing, network-connected outlet interfaces **210** (hereinafter referred to as "smart wall plugs **210**" or "smart outlets"). The smart wall switches **208** may detect ambient lighting conditions, detect room-occupancy states, and control a power and/or dim state of one or more lights. In some instances, smart wall switches **208** may also control a power state or speed of a fan, such as a ceiling fan. The smart wall plugs **210** (or smart outlet plugs) may detect occupancy of a room or enclosure and control supply of power to one or more wall plugs (e.g., such that power is not supplied to the plug if nobody is at home). Further, such smart wall plugs or smart outlet plugs may turn on or off electricity to a connected device based on a command received from an associated cloud-based server system (e.g., **120**, **130-1**, **130-2**).

In some embodiments, one or more smart indoor security cameras may be present such as indoor security camera **272**. Indoor security camera **272** may wirelessly communicate with a cloud server system to capture and record video and audio. Indoor security camera **272** may be able to detect motion, recognize a resident (e.g., via facial detection), and detect the presence of a resident or other person via audio (e.g., detection of a human voice). Indoor and outdoor security cameras may be used to determine when a resident leaves home (for example, an OBM behavior may be a time range during the day when a resident typically leaves home, such as between 8 AM and 6 PM) or returns home. In some embodiments, as previously detailed, this data may be supplemented with location data derived from an electronic device, such as a smartphone, that a resident typically carries when going out. If the smartphone is forgotten by the resident, data from the cameras may be used to determine that the smartphone has been left behind and the resident has left the residence (e.g., a camera detects the resident leaving, the smartphone remains stationary in the residence, and there is no movement detected within the residence for a threshold period of time). When the environment is darkened, indoor security camera **272** may use infrared to detect the presence of a resident and/or other persons in the camera's field-of-view.

In some embodiments, one or more smart outdoor security cameras may be present such as outdoor security camera **276**. Outdoor security camera **276** may wirelessly communicate with a cloud server system to capture and record video and audio and may function similarly to indoor security camera **272**. Outdoor security camera **276** may include weatherproofing to protect against the outdoor environment. Outdoor security camera **276** may be able to detect motion, recognize a resident (e.g., via facial detection), and detect the presence of a resident or other person via audio (e.g., detection of a human voice). At night, outdoor security camera **276** may use infrared to detect the presence of a resident and/or other persons in the camera's field-of-view.

In some embodiments, one or more home assistant devices may be present in the residence, such as home assistant device **274**. Home assistant device **274** may include one or more microphones. Home assistant device **274** may detect and analyze human speech and may be able to detect speech and/or movement by the resident. Clips of captured human speech may be provided to cloud-based host server system **120** for analysis. A response may be provide to home assistant device **274**, which may output the response in the form of spoken audio. Cloud-based host server system **120** may cause one or more other smart-home devices in home environment **200** to perform a function based on a command identified from the received human speech.

Still further, in some embodiments, the smart-home environment **200** of FIG. **2** includes a plurality of intelligent, multi-sensing, network-connected appliances **212** (hereinafter referred to as "smart appliances **212**"), such as refrigerators, stoves and/or ovens, televisions, washers, dryers, lights, stereos, intercom systems, garage-door openers, floor fans, ceiling fans, wall air conditioners, pool heaters, irrigation systems, security systems, and so forth. According to embodiments, the network-connected appliances **212** are made compatible with the smart-home environment by cooperating with the respective manufacturers of the appliances. For example, the appliances can be space heaters, window AC units, motorized duct vents, etc. When plugged in, an appliance can announce itself to the smart-home network, such as by indicating what type of appliance it is, and it can automatically integrate with the controls of the smart-home. Such communication by the appliance to the smart-home can be facilitated by any wired or wireless communication protocols known by those having ordinary skill in the art. The smart-home also can include a variety of non-communicating legacy appliances **240**, such as old conventional washer/dryers, refrigerators, lights, and the like which can be controlled, albeit coarsely (ON/OFF), by virtue of the smart wall plugs **210**. The smart-home environment **200** can further include a variety of partially communicating legacy appliances **242**, such as infrared ("IR") controlled wall air conditioners or other IR-controlled devices, which can be controlled by IR signals provided by the hazard detectors **204** or the smart wall switches **208**.

According to embodiments, the smart thermostats **202**, the hazard detectors **204**, the smart doorbells **206**, the smart wall switches **208**, the smart wall plugs **210**, and other devices of the smart-home environment **200** are modular and can be incorporated into older and new houses. For example, the devices are designed around a modular platform consisting of two basic components: a head unit and a back plate, which is also referred to as a docking station. Multiple configurations of the docking station are provided so as to be compatible with any home, such as older and newer homes. However, all of the docking stations include a standard head-connection arrangement, such that any head unit can be removably attached to any docking station. Thus, in some embodiments, the docking stations are interfaces that serve as physical connections to the structure and the voltage wiring of the homes, and the interchangeable head units contain all of the sensors, processors, user interfaces, the batteries, and other functional components of the devices.

The smart-home environment **200** may also include communication with devices outside of the physical home but within a proximate geographical range of the home. For example, the smart-home environment **200** may include a pool heater monitor **214** that communicates a current pool temperature to other devices within the smart-home environment **200** or receives commands for controlling the pool temperature. Similarly, the smart-home environment **200** may include an irrigation monitor **216** that communicates information regarding irrigation systems within the smart-home environment **200** and/or receives control information for controlling such irrigation systems. According to embodiments, an algorithm is provided for considering the geographic location of the smart-home environment **200**, such as based on the zip code or geographic coordinates of the home. The geographic information is then used to obtain data helpful for determining optimal times for watering; such data may include sun location information, temperature, due point, soil type of the land on which the home is located, etc.

By virtue of network connectivity, one or more of the smart-home devices of FIG. **2** can further allow a user to interact with the device even if the user is not proximate to the device. For example, a user can communicate with a device using a computer (e.g., a desktop computer, laptop computer, or tablet) or other portable electronic device **266** (e.g., a smartphone). A webpage or app can be configured to receive communications from the user and control the device based on the communications and/or to present information about the device's operation to the user. For example, the user can view a current setpoint temperature for a device and adjust it, using a computer. The user can be in the structure during this remote communication or outside the structure.

As discussed, users can control and interact with the smart thermostat, hazard detectors **204**, and other smart devices in the smart-home environment **200** using a network-connected computer or portable electronic device **266**. In some examples, some or all of the occupants (e.g., individuals who live in the home) can register their electronic device **266** with the smart-home environment **200**. Such registration can be made at cloud-based host server system **120** (or whichever cloud-based server system is associated with the entity that provided the smart-home devices to be controlled) to authenticate the occupant and/or the device as being associated with the home and to give permission to the occupant to use the device to control the smart devices in the home. An occupant can use their registered electronic device **266** to remotely control the smart devices of the home, such as when the occupant is at work or on vacation. The occupant may also use their registered device to control the smart devices when the occupant is actually located inside the home, such as when the occupant is sitting on a couch inside the home. It should be appreciated that, instead of or in addition to registering electronic devices **266**, the smart-home environment **200** makes inferences about which individuals live in the home and are therefore occupants and which electronic devices **266** are associated with those individuals. As such, the smart-home environment "learns" who is an occupant and permits the electronic devices **266** associated with those individuals to control the smart devices of the home.

In some embodiments, in addition to containing processing and sensing capabilities, each of the devices **202**, **204**, **206**, **208**, **210**, **212**, **214**, and **216** (collectively referred to as "the smart devices") is capable of data communications and information sharing with any other of the smart devices, as well as to any central server or cloud-computing system or any other device that is network-connected anywhere in the world. The required data communications can be carried out using any of a variety of custom or standard wireless protocols (Wi-Fi, ZigBee, 6LoWPAN, etc.) and/or any of a variety of custom or standard wired protocols (CAT6 Ethernet, HomePlug, etc.)

According to embodiments, all or some of the smart devices can serve as wireless or wired repeaters. For example, a first one of the smart devices can communicate with a second one of the smart devices via a wireless router **260**. The smart devices can further communicate with each other via a connection to a network, such as the Internet **299**. Through the Internet **299**, the smart devices can communicate with a cloud-computing system **264**, which can include one or more centralized or distributed server systems. The cloud-computing system **264** can be associated with a manufacturer, support entity, or service provider associated with the device. For one embodiment, a user may be able to contact customer support using a device itself rather than needing to use other communication means such as a telephone or Internet-connected computer. Further, software updates can be automatically sent from cloud-computing system **264** to devices (e.g., when available, when purchased, or at routine intervals).

According to embodiments, the smart devices combine to create a mesh network of spokesman and low-power nodes in the smart-home environment **200**, where some of the smart devices are "spokesman" nodes and others are "low-powered" nodes. Some of the smart devices in the smart-home environment **200** are battery powered, while others have a regular and reliable power source, such as by connecting to wiring (e.g., to 120V line voltage wires) behind the walls **254** of the smart-home environment. The smart devices that have a regular and reliable power source are referred to as "spokesman" nodes. These nodes are equipped with the capability of using any wireless protocol or manner to facilitate bidirectional communication with any of a variety of other devices in the smart-home environment **200** as well as with the cloud-computing system **264**. On the other hand, the devices that are battery powered are referred to as "low-power" nodes. These nodes tend to be smaller than spokesman nodes and can only communicate using wireless protocols that require very little power, such as Zigbee, 6LoWPAN, etc. Further, some, but not all, low-power nodes are incapable of bidirectional communication. These low-power nodes send messages, but they are unable to "listen". Thus, other devices in the smart-home environment **200**, such as the spokesman nodes, cannot send information to these low-power nodes.

As described, the smart devices serve as low-power and spokesman nodes to create a mesh network in the smart-home environment **200**. Individual low-power nodes in the smart-home environment regularly send out messages regarding what they are sensing, and the other low-powered nodes in the smart-home environment—in addition to sending out their own messages—repeat the messages, thereby causing the messages to travel from node to node (i.e., device to device) throughout the smart-home environment **200**. The spokesman nodes in the smart-home environment **200** are able to "drop down" to low-powered communication protocols to receive these messages, translate the messages to other communication protocols, and send the translated messages to other spokesman nodes and/or cloud-computing system **264**. Thus, the low-powered nodes using low-power communication protocols are able to send messages across the entire smart-home environment **200** as well as over the Internet **263** to cloud-computing system **264**. According to embodiments, the mesh network enables cloud-computing system **264** to regularly receive data from all of the smart devices in the home, make inferences based on the data, and send commands back to one of the smart devices to accomplish some of the smart-home objectives described herein.

As described, the spokesman nodes and some of the low-powered nodes are capable of "listening." Accordingly, users, other devices, and cloud-computing system **264** can communicate controls to the low-powered nodes. For example, a user can use the portable electronic device **266** (e.g., a smartphone) to send commands over the Internet to cloud-computing system **264**, which then relays the commands to the spokesman nodes in the smart-home environment **200**. The spokesman nodes drop down to a low-power protocol to communicate the commands to the low-power nodes throughout the smart-home environment, as well as to other spokesman nodes that did not receive the commands directly from the cloud-computing system **264**.

An example of a low-power node is a smart nightlight **270**. In addition to housing a light source, the smart nightlight **270** houses an occupancy sensor, such as an ultrasonic or passive IR sensor, and an ambient light sensor, such as a photoresistor or a single-pixel sensor that measures light in the room. In some embodiments, the smart nightlight **270** is configured to activate the light source when its ambient light sensor detects that the room is dark and when its occupancy sensor detects that someone is in the room. In other embodiments, the smart nightlight **270** is simply configured to activate the light source when its ambient light sensor detects that the room is dark. Further, according to embodiments, the smart nightlight **270** includes a low-power wireless communication chip (e.g., ZigBee chip) that regularly sends out messages regarding the occupancy of the room and the amount of light in the room, including instantaneous messages coincident with the occupancy sensor detecting the presence of a person in the room. As mentioned above, these messages may be sent wirelessly, using the mesh network, from node to node (i.e., smart device to smart device) within the smart-home environment **200** as well as over the Internet **299** to cloud-computing system **264**.

Other examples of low-powered nodes include battery-operated versions of the hazard detectors **204**. These hazard detectors **204** are often located in an area without access to constant and reliable (e.g., structural) power and, as discussed in detail below, may include any number and type of sensors, such as smoke/fire/heat sensors, carbon monoxide/dioxide sensors, occupancy/motion sensors, ambient light sensors, temperature sensors, humidity sensors, and the like. Furthermore, hazard detectors **204** can send messages that correspond to each of the respective sensors to the other devices and cloud-computing system **264**, such as by using the mesh network as described above.

Examples of spokesman nodes include smart doorbells **206**, smart thermostats **202**, smart wall switches **208**, and smart wall plugs **210**. These devices **202**, **206**, **208**, and **210** are often located near and connected to a reliable power source, and therefore can include more power-consuming components, such as one or more communication chips capable of bidirectional communication in any variety of protocols.

Further included and illustrated in the exemplary smart-home environment **200** of FIG. **2** are service robots **262** each configured to carry out, in an autonomous manner, any of a variety of household tasks. For some embodiments, the service robots **262** can be respectively configured to perform floor sweeping, floor washing, etc. Tasks such as floor sweeping and floor washing can be considered as "away" or "while-away" tasks for purposes of the instant description, as it is generally more desirable for these tasks to be performed when the occupants are not present. For other embodiments, one or more of the service robots **262** are configured to perform tasks such as playing music for an occupant, serving as a localized thermostat for an occupant, serving as a localized air monitor/purifier for an occupant, serving as a localized baby monitor, serving as a localized hazard detector for an occupant, and so forth, it being generally more desirable for such tasks to be carried out in the immediate presence of the human occupant. For purposes of the instant description, such tasks can be considered as "human-facing" or "human-centric" tasks.

As noted in relation to FIG. **1**, not all of the above detailed smart-home devices may be associated with a same manufacturer or entity that provides service for the smart-home device. Therefore, the smart-home devices in smart-home environment **200** may be controlled using different cloud-based server systems. For example, smart-home devices made by a first manufacturer may be controlled via cloud-based host server system **120**, while smart-home devices made by a second manufacturer may be controlled via cloud-based server system **130-1** and smart-home devices made by a third manufacturer may be controlled via cloud-based server system **130-2**. By way of example, three cloud-based server systems, including cloud-based host server system is illustrated in FIG. **2**; in other embodiments, a greater or fewer number of cloud-based server systems may be present.

FIG. **3** illustrates a block diagram of an embodiment of a cloud-based host server system **120**. Cloud-based host server system **120** may communicate directly with various smart-home devices, may interface with other cloud-based server systems (to communicate indirectly with smart-home devices provided by third-party manufacturers), and/or may analyze recorded voice samples received from smart-home devices. Cloud-based host server system **120** may include one or more computer server systems that include components such as network interfaces, processors, communication buses, non-transitory computer readable storage mediums, memories, etc. Various components illustrated as part of cloud-based host server system may be performed using dedicated special-purpose hardware, firmware, or general-purpose hardware executing special-purpose software. It should be understood that the various components of cloud-based host server system **120** may be broken out into a greater number of components or combined into a fewer number of components in different embodiments. Cloud-based host server system **120** may include: smart device control engine **310**; controller communication interface **320**; audio communication interface **330**; semantics engine **340**; assistant engine **350**; smart device cloud API **360**; direct smart-device control API **370**; smart-device control library **380**; and one or more smart device and account databases **390**.

Direct smart device control API **370** may be used to communicate directly with various smart-home devices. The smart devices controlled via direct smart device control API **370** may be manufactured or otherwise distributed by a same entity that operates or has cloud-based host server system **120** operated on its behalf. Through direct smart device control API **370**, data from various smart-home devices may be received and commands may be transmitted to such smart-home devices. Direct smart device control API **370** may format commands appropriately for communication to smart devices and may format received data appropriately for analysis by smart device control engine **310**. In contrast, smart device cloud API **360** may be used to communicate with various smart-home devices through other cloud-based server systems. The smart devices controlled via smart device cloud API **370** may be manufactured or otherwise distributed by a different entity than the entity that operates or has cloud-based host server system **120** operated on its behalf. Through smart device cloud API **360**, data from various third-party smart-home devices may be received and commands may be transmitted to such third-party smart-home devices. Smart device cloud API **360** may format commands appropriately for communication to smart devices and may format received data appropriately for analysis by smart device control engine **310**. Therefore, smart device cloud API **360** may have access to a library of command types and data types available for each third-party smart device.

Controller communication interface **320** may communicate with various smart-home controller devices. Through controller communication interface **320**, requests for a command to be executed by one or more smart-home devices may be received. In some situations, the command may be intended to be executed by one or more smart-home devices directly communicated with by cloud-based host server system **120** via direct smart device control API **370** and one or more smart devices indirectly communicated with via smart device cloud API **360**. Controller communication interface **320** may also be used by smart device control engine **310** to send information about data received from smart devices that communicate with cloud-based host server system **120** via smart device cloud API **360** and direct smart device control API **370**.

Smart device control library **380** may be a library of the various functions that can be performed using smart devices that are directly communicated with via direct smart device control API **370**. When a type of smart device is registered with cloud-based host server system **120**, smart device control library **380** may be accessed to determine the particular functions which the smart device can perform and the particular commands available to be transmitted to smart-device control library **380**.

Smart device and account databases **390** may serve to store indications of smart-home devices that have been registered with cloud-based host server system **120** with a particular user account. Requests to control a smart-home device received via controller communication interface **320** may be required to be linked with the same user account as indicated in smart device an account databases **390** for security purposes. Therefore, a user may only have permission to control the smart-home devices that have been linked to the user's account. In some embodiments, a user name and password is required to access the user account. In other embodiments, additional security measures may be taken, such as biometrics or two-factor identification. One or more smart device an account databases **390** may also store indications of third-party smart-home devices that have been registered with cloud-based host server system **120**.

Audio communication interface **330** may serve to receive an audio clip captured by a smart-home controller, home assistant device, or some other smart-home device through which a user can submit a spoken command. Semantics engine **340** may perform natural language processing to determine what the user said, possibly including: a command; and the device for which the command is intended. Synthesized voice responses may be provided in response to the user via audio communication interface **330**.

Vocal or spoken commands that are not directed to smart-home devices may be processed by assistant engine **350**. For example, such spoken commands may be requesting data from some other source, such as the Internet. As an example, a request may be for weather information, the time, sports scores, a translation, a stock quote, calendar information, etc. semantics engine **340** may route smart-home device requests to smart device control engine **310**.

Smart device control engine **310** may receive commands from controller communication interface **320** and as interpreted by semantics engine **340**. Smart device control engine **310** may access smart device and account databases **390** to determine the smart devices associated with a particular account. Smart-device control engine may determine the appropriate smart-home devices to transmit a command and may transmit the command via smart device cloud API **360** (to the corresponding cloud-based server system that communicates directly with the smart-home device) and via direct smart device control API **370**. Smart device control engine **310** may send smart device identifiers indicative of the smart devices that have been sent a command based on a spoken command to the smart-home controller via controller communication interface **320**. Therefore, smart-home controller may receive information identifying the smart-home devices that were affected by a spoken command.

FIG. **4** illustrates a block diagram of an embodiment of a smart-home controller device. Smart-home controller device **140** may perform multiple functions, including: presenting a graphical interface for a user to view the status of and control multiple smart-home devices (of which some of the smart-home devices may be controlled via different entities' cloud-based server systems); and receive and send vocal commands spoken by a user. Smart-home control device **140** may be a computerized device that includes components such as wireless (and/or wired) network interfaces, processors, communication buses, non-transitory computer readable storage mediums, memories, etc. Various components illustrated as part of smart-home controller device **140** may be performed using dedicated special-purpose hardware, firmware, or general-purpose hardware executing special-purpose software. It should be understood that the various components of smart-home controller device **140** may be broken out into a greater number of components or combined into a fewer number of components in different embodiments. Smart-home controller device **140** may include: voice-based assistant engine **480** and smart-home control application **410**, which may be executed by underlying processing hardware of smart-home controller device **140**. In some embodiments, smart-home control application **410** is downloaded onto smart-home controller device **140** from an application store. In other embodiments, smart-home controller device **140** may be purpose-built to have the functionality of smart-home control application **410**. Voice-based assistant engine **480** may be triggered by a user speaking a key phrase or a user providing input to initiate listening. The user may then speak a vocal command that is passed to the cloud-based host server system for analysis and execution. A synthesized speech response may be output by voice-based assistant engine **480** based on data received in response to the vocal command passed to the cloud-based server system. Voice-based assistant engine **480** may execute regardless of whether smart-home control application **410** is being executed. In some embodiments, voice-based assistant engine **480** is incorporated as part of an operating system (OS) being executed by smart-home controller device **140**.

Smart-home control application **410** may include: smart-home control interface creation engine **420**; smart-home device registration engine **430**; customized controller interface generation engine **440**; voice-based assistant engine **450**; function-specific term database **460**; and smart-home device function database **470**. Smart-home device registration engine **430** may receive information about a new smart-home device that is the registered with smart-home control application **410** and cloud-based host server system **120**, regardless of whether the smart-home device is directly controlled by cloud-based host server system or controlled via another cloud-based server system. As part of the registration process, a user may be requested to provide information about the smart-home device. For example, the user may be requested to provide a name for the smart-home device. The user may typically name the smart-home device based on the function performed by the smart-home device and the location. For example, a user may select a name such as "kitchen light" to describe a smart light in the kitchen. As another example, "kitchen light" may be used for a smart outlet plug in the kitchen to which a conventional light is connected. Registration data received by smart-home device registration engine **430** may be stored locally and may also be transmitted to cloud-based host server system **120**. Functions that can be performed by the smart-home device may be discovered during the smart-home device registration process. Indications of such functions may be stored to smart-home device function database **470** in association with an identifier of the smart-home device. Indications of the available functions may be retrieved from the cloud-based host computer system. For third-party smart-home devices, the cloud-based host computer system may request and retrieve indications of such functions from the cloud-based server system operated by the entity that provided the third-party smart-home device. Alternatively, the cloud-based host server system may maintain a database of available functions of third-party smart-home devices.

When the user provides a name for a smart-home device as part of the registration process, if the ultimate function of the smart-home device is not inherently determined by the type of smart device itself, function-specific term database **460** may be used to determine the intended use of a smart-home device from the user's name for the smart-home device. For example, a smart outlet plug can be used to control a variety of types of devices, such as a legacy appliance, a light, electronics, etc. A user may be expected to name the smart outlet plug something related to the device it is controlling, such as "bedroom window fan" or "dining room light." Function specific term database may store a variety of terms that are linked to a specific category of smart-home device. Table 1 provides some examples of terms and the associated smart-home device categories.

TABLE 1

Function-Specific Term	Smart-home Device Category
Light	lighting
Lamp	Lighting

TABLE 1-continued

Function-Specific Term	Smart-home Device Category
Bulb	Lighting
Speaker	Music/sound output
Stereo	Music/sound output

If, during the registration process, a user provides a function specific term in the name of the smart-home device that performs an ambiguous function, based on the function specific term being present, smart-home device registration engine **430** may classify the smart-home device as being part of the smart-home device category mapped to the function specific term. For example, a smart outlet plug that has been named “dining room lamp” may be classified as part of the lighting smart-home device category. While FIG. **4** shows function specific term database **460** as part of smart-home control application **410**, in other embodiments, function-specific term database **460** may be remotely located and may be accessible by smart-home device registration engine **430**. For instance, function-specific term database **460** may be stored by cloud-based host server system **120**.

Voice-based assistant engine **450** may function similarly to voice-based assistant engine **480**, but may be incorporated as part of smart-home control application **410** and may specifically be used for providing smart-home related commands. Voice-based assistant engine **450** may capture an audio clip via a microphone of the smart-home controller device **140** in response to a user saying a trigger phrase or providing some form of user input indicating that the user desires to provide a voice-based command. The captured audio may be transmitted from voice-based assistant engine **450** to audio communication interface **330** of cloud-based host server system **120**. The audio may be analyzed at the cloud-based host server system rather than at smart-home controller device **140**. In response to sending the captured audio, voice-based assistant engine **450** may receive a response from the cloud-based host computer system indicating the smart-home identifiers of the smart-home devices that were sent to command based on a spoken command in the captured audio. The response may also indicate the updated state of the smart-home devices to which the command was sent.

Smart-home control interface creation engine **420** may create graphical interfaces that can be interacted with by user via a touchscreen or other form of user input device and is presented by smart-home controller device **140** or otherwise output for presentation (e.g., to a separate monitor). Smart-home control interface creation engine **420** may create an output user interfaces such as those detailed in relation to FIGS. **5-8**. Smart-home control interface creation engine **420** may create individual tiles or coins that control both smart-home devices via direct smart device control API **370** and third-party smart-home devices via smart device cloud API **360**.

Smart-home device function database **470** may store an indication of each smart-home device that is controlled via smart-home control application **410** and the functions that can be performed at each of the smart-home devices. For example, many smart-home devices may be able to be turned on and off, some smart lights may be able to be dimmed, smart thermostats may be able to receive a temperature setting, etc. smart-home device function database **470** may be accessed by smart-home control interface creation engine **420** to determine the functions that can be performed by a particular smart-home device. Smart-home control interface creation engine **420** may access smart-home device function database **470** to determine what functions are in common across particular smart-home devices such that a single control can be presented that controls multiple smart-home devices, even if some of such smart-home devices are provided by different entities and are controlled via different cloud-based server systems.

Customized controller interface generation engine **440** may be used in conjunction with voice-based assistant engine **450**. In response to a voice command, voice-based assistant engine **450** may receive smart device identifiers for the smart devices affected by a spoken command. When these smart device identifiers are received, the identifiers may be passed to customized controller interface generation engine **440**. Customized controller interface generation engine **440** may access smart-home device function database **470** to determine one or more functions that each of the smart-home devices have in common. A graphical control may be created that allows a user to control the same group of smart-home devices that was indicated in the spoken command. This control may include at least one common function across each of the smart-home devices. The control may also include one or more functions that are specific to a subset of the smart-home devices that was indicated in the spoken command. Therefore, customized controller interface generation engine **440** may generate a graphical interface after a spoken command has been submitted via voice-based assistant engine **450** and may allow a user to provide user input in the form of touch or some other form of on-screen control for the same set of smart-home devices that was previously controlled via a vocal command. Additionally or alternatively, customized controller interface generation engine **440** may be used with voice-based assistant engine **480**.

FIGS. **5-8** represent various control interfaces that may be presented by smart-home controller devices **140**. In the illustrated embodiments, a smartphone **142** is used as an example of the smart-home controller device. In some embodiments, the control interfaces are presented using a touchscreen, in other embodiments, a separate user input device, such as a mouse is used to interact with the control interfaces. FIGS. **5-8** represent control interfaces that may be output by smart-home control application **410** for presentation by smart-home controller device **140** or a display device connected with smart-home controller device **140**. FIG. **5** illustrates an embodiment of a control interface **500** that may be presented by a smart-home controller device, such as smartphone **142**.

Top interface region **510** is devoted to whole-home smart-device controls. Top interface region **510** is labeled with location **505** that is indicative of the entire structure or other form of place where the smart-home devices are installed. Controls with top interface region **510** can include: off element **511**; on element **512**; play element **513**; thermostat element **514**; camera element **515**; add element **516**; and settings element **517**.

The controls indicated by elements **511-516** pertain to smart-home devices controlled directly via the cloud-based host server system and third-party smart-home devices controlled by the cloud-based host server system via another cloud-based server system operated by another entity. Off element **511**, when selected by a user, may trigger the smart-home control application to instruct the cloud-based host server system to turn off all lighting-related smart-home devices at location **505** that are located in lower location subcategories. (In other embodiments, when off element **511** is selected, a choice may be presented as to whether to turn off lighting-related devices for a given room or within the entire structure.) On element **512**, when selected by a user, may trigger the smart-home control application to instruct the cloud-based host server system to turn on all lighting-related smart-home devices at location **505** that are located in lower location subcategories. (In other embodiments, when on element **512** is selected, a choice may be presented as to whether to turn off lighting-related devices for a given room or within the entire structure.) Play element **513**, when selected by a user, may trigger the smart-home control application to instruct the cloud-based host server system to play or cease playing sound via eligible smart-home devices at location **505** that are located in lower location subcategories. Thermostat element **514**, when selected by a user, may trigger the smart-home control application to instruct the cloud-based host server system to adjust the temperature or adjust an away/home setting at all smart thermostats installed in location **505** that are located in lower location subcategories. Camera element **515**, when selected by a user, may trigger the smart-home control application to instruct the cloud-based host server system to stream video feeds to the smart-home control device for presentation from all cameras installed at location **505** that are located in lower location subcategories. Add element **516** may allow an additional function to be added to top interface region **510** based on one or more smart-home devices present in lower location subcategories. Alternatively, add element **516** may be used to link or register a new device. Settings element **517**, when selected by a user, may trigger the smart-home control application to present various settings that a user is permitted to adjust for controlling smart-home devices located in lower location subcategories.

Based on received registration information received from a user via smart-home device registration engine **430**, smart-home devices may be classified into one or more location subcategories. Location subcategories may be presented within control interface **500** at a lower location than top interface region **510**. A first location subcategory presenting in control interface **500** is location subcategory **520** that has location **521** of "Attic." A user may specify location **521** and indicate which smart-home devices are installed at location **521**. All smart-home devices indicated as being at location **521** are included in location subcategory **520**. In the illustrated embodiment, three smart-home devices (of which at least some may be third-party smart-home devices) is indicated by smart-home device count **522**, which is presented near location **521**. In the illustrated embodiment, two lights are present at location **521**, therefore control element **524**, which may also be referred to as a control "coin" or "tile" is presented that includes: 1) a graphic that represents a group of the type of smart-home device and may be selected to present control elements for the individual lights; 2) a title of the group of lights ("Attic lights"), and control elements that allow the group of lights to be controlled using a single user input (by a user selecting "off" or "on"). Additionally, location subcategory **520** may include a smart-home device (in this case, a home assistant device) that can be used as a speaker. Control element **523** include: 1) a graphic element that represents the type of smart-home device; and 2) a user-assigned title of the smart-home device. Additional location subcategories may be presented by control interface **500** by a user scrolling down, such as seen in the embodiment of FIG. **6**.

Additionally present in control interface **500** is spoken command control element **530**. By a user selecting audio command control element **530**, voice-based assistant engine **450** may be activated to capture audio using a microphone of smart-home controller device **140**. This audio may be transmitted to cloud-based host server system **120** for analysis. When a user scrolls up or down within control interface **500**, spoken command control element **530** remains fixed on control interface **500** such that while smart-home control application **410** is open, the user has an on-screen option to provide a spoken command to control the smart-home devices.

FIG. **6** illustrates an embodiment of a control interface **600** that may be presented by a smart-home controller device that includes grouped smart-home devices based on location. In control interface **600**, location subcategories **610** and **620** are presented. Location subcategory **610** corresponds to location **611** and has four devices as indicated by smart-home device count **612**. If a user selects location **611**, interface **700** of FIG. **7** may be presented. In the illustrated embodiment, three lights are present at location **611**, therefore control element **613** may include: 1) a graphic that represents a group of the type of smart-home device and may be selected to present control elements for the individual lights and/or the current state of the group of lights; 2) a title of the group of lights ("Dining room lights"), and control elements that allow the group of lights to be collectively controlled using a single user input (by a user selecting "off" or "on"). Control element **614** may allow a user to pair a speaker at location **611** with a source (e.g., the smart-home control device) for output of sound.

Location subcategory **620** which corresponds to location **621** only includes one smart-device as indicated by smart-device count **622**. Control element **623** visually represents a smart door lock and indicates the current state of the door lock (unlocked).

FIG. **7** illustrates an embodiment of a control interface **700** that may be presented by smart-home controller device **140** for a particular location subcategory. Control interface **700** may be presented when a user selects location **611** in control interface **600**. In interface **700**, control elements are presented for: 1) groups of smart-home devices of the same type; and 2) each individual smart-home device. Control element **613** and control element **614** may function as detailed in relation to FIG. **6**. Additionally, control elements **701**, **702**, and **703** are presented. Control element **701** may correspond to a smart light that is directly controlled by cloud-based host server system **120** via direct smart device control API **370**. Control element **702** may correspond to a smart light made or distributed by another entity that is controlled by cloud-based host server system **120** via smart-device cloud API **360** and the entity's cloud-based server system. The graphical representations presented as part of control elements **701** and **702** may differ to illustrate the that each smart light is from a different manufacturer or entity. However, both of these smart lights can be controlled using the single control of control element **613**.

Control element **703** comprises a graphical representation of a smart outlet plug. This smart outlet plug may have been grouped as part of the smart lights of location **611** on the basis of the name for the smart outlet plug provided for a user including the word "lamp." The graphical representation may remain as a smart outlet plug. In some embodiments the smart outlet plug is included as part of location **611** based on the word "dining" being used as part of the user-provided name.

FIG. **8** illustrates an embodiment of a control interface **800** that may be presented by smart-home controller device **140** that can be used to control multiple smart-home devices. Control interface **800** may be presented when control element **613** in selected from control interface **600** or interface

700. Of control interface **800**, at least one control element controls all smart-elements within the selected group **805**. Smart device count **806** indicates the number of smart devices that are controlled by the at least one control element that controls all smart-elements within the selected group **805**. Control interface **800** may allow a single user input to control multiple smart lights which include smart lights controlled directly by cloud-based host server system and smart lights controlled via another cloud-based server system. Power control element **801** may control power for every smart light (that support a power control function) at location **611** and its state may be indicated by state **802**. Therefore, with one instance of user input, a user can turn on or off all lights within the dining room, regardless of the manufacturer or entity from which the smart light was obtained. Dimmer control element **803** may allow for controlling the dim level of all smart-lights at location **611** that support the feature of dimming. Therefore, while power control element **801** controls all smart lights within selected group **805**, control element **803** only controls smart lights that support the feature (which may be all or some of the smart lights within selected group **805**). For example, a light controlled by a smart outlet plug, such as control element **703**, may not be able to control brightness.

Various methods may be performed using the systems and control interfaces of FIGS. **1**-**8**. FIG. **9** illustrates an embodiment of a method **900** for integrating control of multiple cloud-based smart-home devices. Steps of method **900** may be performed using smart-home controller device **140**, which may be executing smart-home control application **410**. Smart-home controller device **140** may be functioning as part of system **100** and smart-home environment **200** of FIGS. **1** and **2**, respectively.

At block **905**, the smart-home control application **410** may receive registration for a first smart-home device by the smart-home control application or the registration information may be received by the cloud-based host server system. This smart-home device may be associated with the manufacture or entity that operates cloud-based host server system **120**. Therefore, the first smart-home device may be controlled via direct smart device control API **370**. As part of the registration process, the user may provide a user-defined name for the smart-home device. The user may, additionally or alternatively, provide a location that indicates a room or other area within or at a structure where the first smart-home device is located or will be located. In some embodiments, the location is selected from a predefined list of locations available within smart-home control application **410**. For example, smart-home control application **410** may list the names of rooms that are commonly found within a residential home. In other embodiments, a user may specify a custom name for the location. In still other embodiments, the location at which the smart-home device was installed may be inferred from a name for the smart-home device defined by the user. For example, if the name includes the word "dining" it may be determined that the first smart-home device is installed in a dining room. The registration information may also include an identifier of the smart-home device which can be used to register the smart-home device with the cloud-based host computer system and, if necessary, with the cloud-based server system with which the cloud-based host computer system can communicate. In some embodiments, the cloud-based host computer system may return in indication of the type of smart device being registered and functions that the smart-home device is capable of performing. For the purposes of the remainder of method **900**, it is assumed that the first smart-home device was registered with the cloud-based host computer system and the first smart-home device communicates directly with the cloud-based host computer system.

At block **910**, registration information for a second smart-home device may be received by the smart-home control application or the registration information may be received by the cloud-based host server system. Registration for the second smart-home device may proceed similarly to block **910**. In the illustrated embodiment of method **900**, the second smart-home device is controlled via a cloud-based server system distinct from the cloud-based host server system. Therefore, when the second smart-home devices registered with the cloud-based host server system, the cloud-based host server system may provide at least some of the registration information to the cloud-based server system. In response, the cloud-based server system may provide indications of functions that may be performed by the second smart-home device. For the purpose of method **900**, the first smart-home device and the second smart-home device are of a same type (e.g., are both types of smart lights) and have at least one function in common, such as both devices can be turned on and off (e.g., a "power function"). However, the first smart-home device in the second smart-home device are provided by different manufacturers or entities and are controlled using different cloud-based systems.

At block **915**, a determination may be made by the smart-home control application **410** that a common function exists for the first smart-home device and the second smart-home device. In some embodiments, block **915** may be performed by the cloud-based host server system. This determination may be made by comparing a list of available functions for the first smart-home device determined during the registration of block **905** and the list of available functions for the second smart-home device determined during the registration of block **910**. Block **915** may only be performed if the first smart-home device and the second smart-home device are determined to be of the same type (e.g., lights, speakers, locks, fans, thermostats, etc.). Block **915** may further include determining that the first smart-home device in the second smart-home device are being or have been installed at the same location (e.g., within a same room).

At block **920**, the first smart-home device and the second smart-home device may be assigned to a common operating characteristic group by the smart-home control application. In some embodiments, block **920** may be performed by the cloud-based host server system. In some embodiments, the first smart-home device in the second smart-home device may be assigned to multiple common operating characteristic groups. For example, a first common operating characteristic group may be created for all smart-home devices of the same type within a structure. For example, such an arrangement may allow a control element such as element **511** and element **512** to control all smart lights within a structure. Alternatively or additionally, a second common operating characteristic group may be created for all smart-home devices of the same type within a particular location at a structure. For example, such an arrangement may allow for a control element such as control element **613** to control all smart lights in the dining room, regardless of the entity that provided the smart light.

At block **925**, the smart-home control application may provide a control element that includes a control for allowing a user to control the common function determined at block **915**. Therefore, by a user providing a single input to the control, the user can adjust a function or setting of multiple smart-home devices. In the example of method **900**, the single input may adjust a function at the first smart-home device in the second smart-home device, which are controlled via different cloud-based systems.

At block **930**, user input may be received to the control of the control element that allows a user to control the common function. A single instance of user input may be received at block **930**, such as a single touch or a single click. As an example, referring to control element **613**, a user may touch the "on" control to turn three smart lights on or the "off" control to turn three smart lights off. In this example, "on" and "off" would be common functions across the multiple smart lights. At block **935**, the smart-home control application may cause the smart-home controller device to transmit a message to the cloud-based host system that indicates smart-home device identifiers of the first smart-home device in the second smart-home device (and any other smart-home devices that are of the same type, and, possibly location). The message may also include an indication of the command or function to be performed by each of the smart-home devices. In some embodiments, the user input may be provided to a device other than the device executing the smart-home control application. For instance, the user input may be provided to the cloud-based host server system.

At block **940**, in response to the received message, the cloud-based host server system may cause the first smart-home device to perform the commander function. This may be performed by the cloud-based host server system communicating with the first smart-home device via direct smart device control API **370**. Additionally, in response the received message, the cloud-based host server system may send an indication of the second smart-home device identifier and the command or function to a cloud-based server system that communicates with the second smart-home device. The cloud-based server system may then instruct the second smart-home device to perform the command or function. Therefore, by user providing a single user input at block **930**, smart devices controlled via different cloud-based server systems may each be controlled. In some embodiments, the cloud-based server system may provide an acknowledgment as to whether the second smart-home device performed the command or function to the cloud-based host system. Additionally or alternatively, the cloud-based host server system may provide an acknowledgment to the smart-home control application indicating whether the function or command has been performed by the first smart-home device and/or the second smart-home device. Such an acknowledgment may permit presentation of an interface of the smart-home control application to be updated to reflect the current state of the first and second smart-home devices.

FIG. **10** illustrates an embodiment of a method **1000** for determining how to group a smart-home device. Steps of method **1000** may be performed using smart-home controller device **140**, which may be executing smart-home control application **410**. Smart-home controller device **140** may be functioning as part of system **100** and smart-home environment **200** of FIGS. **1** and **2**, respectively.

At block **1010**, registration information may be received for a first smart-home device, as in block **905** of FIG. **9**. As part of the registration process, the user may provide a user-defined name for the smart-home device. The user may, additionally or alternatively, provide a location that indicates a room or other area within or at a structure where the first smart-home device is located or will be located. In some embodiments, the location is selected from a predefined list of locations available within smart-home control application **410**. For example, smart-home control application **410** may list the names of rooms that are commonly found within a residential home. In other embodiments, a user may specify a custom name for the location. The first smart-home device may communicate directly with the cloud-based host server system or through another cloud-based server system if it is a smart-home device manufactured or distributed by a third-party.

At block **1020**, a determination may be made at the first smart-home devices eligible to perform multiple of functions. Block **1020** may be performed by the smart-home application or by the cloud-based host server system. Which functions may be performed by the first smart-home device may be based on inquiry being made to the cloud-based host server system or a cloud-based server system which communicates with the cloud-based host server system. In other embodiments, a database may be maintained locally by the smart-home control application that indicates the available functions of various types of smart-home devices. An identifier of the first smart-home device may be used by either the smart-home host server system or the smart-home control application to perform a lookup within the database. As an example, a smart outlet plug may be considered one type of smart-home device that is eligible to perform multiple types of functions. The smart-home outlet plug may be able to turn on and off a device that is plugged into the smart-home all the plug. Therefore, the function of first smart-home device may be considered to be the function of whatever devices plugged into the smart-home outlet plug. Typical devices that may be plugged into the smart-home all the plug may be a light, a fan, or a small appliance (e.g., coffee maker, toaster oven, etc.).

Based on block **1020** being determined that the first smart-home devices eligible to perform multiple types of functions, a comparison may be made at block **1030** to compare the user-defined name received at block **1010** with the database of function specific terms. The database a function specific terms may be similar to those presented in Table 1. The comparison of block **1030** may be performed locally by the smart-home controller application or may be performed remotely by the cloud-based host server system. Block **1030** may be performed by the smart-home application or by the cloud-based host server system.

At block **1040**, based on the comparison of block **1030**, a determination may be made as far as the intended function of the first smart-home device. Block **1040** may be performed by the smart-home application or by the cloud-based host server system. Therefore, while the first smart-home device may be of a first type, the way the first smart-home devices being used may be more accurately described based on its intended function. As an example of this consider a smart-home outlet plug that is connected with a light. From a user's point of view, the first smart-home device is causing the light to function as a smart light. Therefore, the smart-home outlet plug may have its intended function be determined as lighting.

At block **1050**, the first smart-home device may be grouped with a second smart-home device based on the intended function of the first smart-home device matching the function of the second smart-home device by the smart-home application or by the cloud-based host server system. Returning to the previous example, the smart outlet plug may be grouped with a smart light (the second smart-home device) due to the intended function of the first smart-home device matching the function of the smart light. Another example of this can be seen in FIG. **6**, in which control element **703** controls a smart outlet plug as part of a group of lights that includes control element **701** and control element **702**. The intended function of the smart outlet plug is determined to match a function of the other lamps.

Additionally, grouping may be further based on location within a structure. For example, a smart outlet plug may only be grouped with a second smart-home device that matches the smart outlet plug's intended function if the smart-home outlet plug in the second smart-home device are indicated as being present at a same location. In some embodiments, similar to the user defined name being analyzed for an intended function using a database of function specific terms, the user defined name may be analyzed for an intended location using a database of location names. Therefore, rather than a user selecting a location from a list or otherwise providing a specific location, the user defined name may be used to determine the location.

As detailed in relation to FIG. **4**, a voice-based assistant engine may be used to control smart-home devices. Such a voice-based assistant engine may function as part of smart-home control application **410** or be separately executed by smart-home controller device **140**, such as by or part of an operating system of smart-home controller device **140**. FIGS. **11**-**14** illustrate embodiments of interfaces that may be presented using a smart-home controller device. In the illustrated embodiments, a smartphone **142** is used as an example of the smart-home controller device.

FIG. **11** illustrates an embodiment of an audio command control **1100** that may be presented when a smart-home controller device is ready to capture an audio sample. Audio command control **1100** may be presented when a user has selected spoken command control element **530**. Alternatively, a user may speak a trigger phrase (e.g., "OK Home Assistant") or provide some other form of input (e.g., squeezing the sides of the smart-home controller device). Once audio command control **1100** is active, one or more microphones of the smart-home controller device may be being used to capture audio from the environment.

FIG. **12** illustrates an embodiment of an audio command control **1200** that may be presented while capturing an audio sample. When a user starts speaking a command, as the audio is received, a voice recognition process may be performed at the cloud-based server system. Text that represents the speech (representative text **1201**) may be presented via audio command control **1200**. In other embodiments, the speech is analyzed locally by the voice-based assistant engine rather than being sent to the cloud for analysis. Presentation of the text of the words being recognized from the spoken command allows a user to confirm the spoken command is being interpreted correctly. In the example of FIG. **12**, a user has spoken "turn off the dining room lights and turn off the living rooms lights." Therefore, with this single command, the user is trying to collectively control both smart lights in dining room and living room.

FIG. **13** illustrates an embodiment of an interface **1300** that includes a custom interface control **1310** that can control a common function across smart-home devices indicated in the audio sample. Interface **1300** is presented in response to the command indicated by representative text **1201** being performed. Custom location group **1301** indicates the group of locations indicated by the user in the spoken command. Smart device count **1302** indicates the total number of smart-home devices that were controlled based on the spoken command indicated by representative text **1201**. Control element **1311** may allow for control of all smart-home devices indicated by smart device count **1302** to perform a function that is common to all of the smart-home devices that were controlled based on the spoken command. In this case, control element **1311** can receive user input to adjust whether the previously-controlled smart-home devices are turned on or off.

In some embodiments, one or more additional control elements may be present. In this embodiment, control element **1312** is present, which is a dimmer control that allows for some or all of the controlled smart-home devices to have their brightness controlled. By control element **1312** being present, this is an indication that at least one of the smart-home devices controlled via the spoken command can have its brightness controlled. In some embodiments, control element **1312** may only be presented and available if this function is available across all smart-home devices controlled by the spoken command indicated by representative text **1201**.

FIG. **14** illustrates an embodiment of a custom interface **1400** allows a user to view video streams indicated in the audio sample. Custom interface **1400** may presented in response to a user providing an audio command of "show all of my video feeds." This command may result in each video feed that is received by the smart-home host server system being output for presentation to custom interface **1400**. Title **1401** may be representative of the custom group of videos that the user indicated in the voice command. Device count **1402** indicates the number of video streams linked to the user's account. Each video stream (**1403**, **1404**, **1405**) may be presented for viewing.

FIG. **15** illustrates an embodiment of a method **1500** for using captured voice to generate an interactive custom interface controller. Steps of method **1500** may be performed using smart-home controller device **140**, which may be executing smart-home control application **410**. Smart-home controller device **140** may be functioning as part of system **100** and smart-home environment **200** of FIGS. **1** and **2**, respectively.

At block **1505**, a local recording may be captured using the smart-home controller device that includes a spoken command intended by the user to be directed to multiple smart-home devices. Alternatively, the recording may be captured using some other device. These multiple smart-home devices may be spread among multiple locations or concentrated at a particular location. These multiple smart-home devices may also represent a subset of smart-home devices located at a particular location. The multiple smart-home devices may be of different types or of the same type. The multiple smart-home devices may include smart-home devices that are directly controlled by the cloud-based host server system and may include smart-home devices that are controlled by the cloud-based host server system via another cloud-based server system. Therefore, the user has considerable flexibility in the smart-home devices that can be controlled using a single spoken command. As previously detailed the smart-home controller device may be ready to receive a spoken command when an interface similar to that of audio command control **1100** is presented.

At block **1510**, the vocal recording may be transmitted to the cloud-based host server system to analyze the contents and meaning. In other embodiments, the vocal recording may be analyzed locally. The vocal recorded captured at block **1505** may be transmitted as the audio is received to the cloud-based host server system to allow for analysis as the voice recording is still being received. At block **1515**, the vocal recording may be translated from audio into text by the cloud-based host server system. In other embodiments, the vocal recording may be analyzed locally by the smart-home controller device. A semantics engine may analyze the intended meaning of the text including the command to be performed the smart-home devices at which the command is intended to be executed.

At block **1520**, the command may be transmitted to the smart-home devices determined at block **1515**. The command may be transmitted to smart-home devices with which the cloud-based host server system directly communicates and smart-home devices with which the smart-home host server system communicates with via another cloud-based server system. In some embodiments, the smart-home controller device may communicate directly with the cloud-based host server systems associated with various smart-home devices. At block **1525**, a response may be transmitted to the smart-home controller device at which the vocal recording was initially received. The response may indicate smart-home device identifiers indicative of each smart-home device to which the command was transmitted by the smart-home host server system. Therefore, if the smart-home host server system sent the command to six smart-home devices, six smart-home identifiers would be transmitted to the smart-home controller device. In some embodiments, rather than the smart-home devices being controlled via cloud-based server systems, the smart-home controller device may communicate with and control the smart-home devices directly (e.g., via a direct wireless communication method or via a mesh network).

At block **1530**, the smart-home device identifiers may be received and then analyzed by the smart-home control application being executed by the smart-home controller device. Each of the smart-home device identifiers may be used to identify the associated smart-home devices and access a smart-home device function database to determine what one or more functions (or intended functions) each smart-home device is capable of. This analysis may include determining if there are one or more functions in common across all of the smart-home devices that were controlled via the spoken command this analysis may also include determining if there one or more functions in common across some of the smart-home devices that were controlled via the spoken command. (In other embodiments, such a function may be present at only one of the smart-home devices that were previously controlled via the voice-based command.) In some embodiments, rather than the analysis of block **1530** being performed at the application executed by the smart-home controller device, the analysis may be performed by the cloud-based host server system to determine what functions each device is capable of and determine if one or more functions are in common across the smart-home devices.

If there is at least one function that is in common across all of the smart-home devices that were controlled via the spoken command, a custom interface controller may be generated at block **1535** that includes one or more controls that allow for user control of the one or more common functions of the smart-home devices that were the subject of the spoken command. This custom interface controller may not have been available to a user via the application prior to block **1535**. For example, no interface controller may be available that controls lights in two distinct rooms. Therefore, block **1530** results in the creation of an interface at block **1535** that otherwise would not exist or be available. The custom interface controller may also include one or more controls that control one or more functions that are in common across some of the smart-home devices that were controlled via the spoken command. Therefore, the presented custom interface controller allows a user to provide additional input to control the smart-home devices that were previously controlled via a voice-based command. In some embodiments, even if there is not one function in common across the smart-home devices indicated by the spoken command, a custom interface controller may be generated that includes a control for a majority of the smart-home devices controlled via the spoken command.

Using the custom interface controller generated at block **1535** that is being output for presentation, a user may provide user input (e.g., via touch) to further control all or some of the smart-home devices previously controlled via the spoken command. In response to such further input, the smart-home control application may determine one or more commands to be performed by the smart-home devices. The command, along with the smart-home device identifiers indicative of the smart-home devices that are to perform the command may be transmitted to the cloud-based host server system for execution. The cloud-based host server system then may control one or more smart-home devices directly and/or may communicate with one or more other cloud-based server systems to control the corresponding smart-home devices.

The methods, systems, and devices discussed above are examples. Various configurations may omit, substitute, or add various procedures or components as appropriate. For instance, in alternative configurations, the methods may be performed in an order different from that described, and/or various stages may be added, omitted, and/or combined. Also, features described with respect to certain configurations may be combined in various other configurations. Different aspects and elements of the configurations may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples and do not limit the scope of the disclosure or claims.

Specific details are given in the description to provide a thorough understanding of example configurations (including implementations). However, configurations may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the configurations. This description provides example configurations only, and does not limit the scope, applicability, or configurations of the claims. Rather, the preceding description of the configurations will provide those skilled in the art with an enabling description for implementing described techniques. Various changes may be made in the function and arrangement of elements without departing from the spirit or scope of the disclosure.

Also, configurations may be described as a process which is depicted as a flow diagram or block diagram. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure. Furthermore, examples of the methods may be implemented by hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof. When implemented in software, firmware, middleware, or microcode, the program code or code segments to perform the necessary tasks may be stored in a non-transitory computer-readable medium such as a storage medium. Processors may perform the described tasks.

Having described several example configurations, various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the disclosure. For example, the above elements may be components of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after the above elements are considered.

What is claimed is:

1. A method for integrating control of multiple cloud-based smart-home devices, the method comprising:

receiving, by an application being executed by a mobile device, a first set of smart-home device registration information for a first smart-home device, wherein the first set of smart-home device registration information comprises a location within a structure;

receiving, by the application being executed by the mobile device, a second set of smart-home device registration information for a second smart-home device, wherein the second set of smart-home device registration information comprises the location;

determining, by the application being executed by the mobile device, that the first smart-home device and the second smart-home device share a common function and are both linked to the location;

assigning the first smart-home device and the second smart-home device to a common operating characteristic group based on the common function being shared by the first smart-home device and the second smart-home device and both being located at the location; and

presenting, within a touchscreen control interface of the application being executed by the mobile device, a first selectable control element and a second selectable control element that control smart-home devices within the common operating characteristic group, wherein:

the first selectable control element and the second selectable control element are presented concurrently as part of the touchscreen control interface of the application; the touchscreen control interface is configured such that the second selectable control element controls all smart-home devices having the common function located within the structure regardless of location within the structure; and

the touchscreen control interface is configured such that the first selectable control element controls only smart-home devices having the common function at the location.

2. The method of claim 1, wherein:

a first third-party manufacturer that provides the first smart-home device operates a first cloud-based server system used to control the first smart-home device; and

a second third-party manufacturer that provides the second smart-home device operates a second cloud-based server system used to control the second smart-home device, the second third-party manufacturer being distinct from the first third-party manufacturer and the second cloud-based server system being distinct from the first cloud-based server system.

3. The method of claim 2, further comprising:

causing, by the application, a message to be transmitted to a cloud-based host system for the first smart-home device, wherein the cloud-based host system controls the common function of the first smart-home device via communication with the first cloud-based server system.

4. The method of claim 3, wherein the message transmitted to the cloud-based host system is further for the second smart-home device, wherein the cloud-based host system controls the common function of the second smart-home device via communication with the second cloud-based server system.

5. The method of claim 4, wherein the location is a room within the structure.

6. The method of claim 5, wherein a smart device count is presented concurrently with the first selectable control element, the smart device count indicating a total number of devices controlled by the first selectable control element.

7. The method of claim 6, wherein the second selectable control element is presented in the application as part of a top interface region.

8. The method of claim 1, further comprising:

presenting, by the application, concurrently with the first selectable control element, a third selectable control element and a fourth selectable control element, wherein the third selectable control element controls only the first smart-home device and the fourth selectable control element controls only the second smart-home device.

9. The method of claim 1, further comprising:

presenting, by the application, concurrently with the first selectable control element, a state indication for the common operating characteristic group.

10. A system for integrating control of multiple cloud-based smart-home devices, the system comprising:

an application executed by a mobile device, wherein the application causes one or more processors of the mobile device to:

receive a first set of smart-home device registration information for a first smart-home device, wherein the first set of smart-home device registration information comprises a location;

receive a second set of smart-home device registration information for a second smart-home device, wherein the second set of smart-home device registration information comprises the location;

determine that the first smart-home device and the second smart-home device share a common function and are both at the location;

assign the first smart-home device and the second smart-home device to a common operating characteristic group based on the common function being shared by the first smart-home device and the second smart-home device and both being located at the location; and

present a first selectable control element and a second selectable control element that control smart-home devices within the common operating characteristic group, wherein:

the first selectable control element and the second selectable control element are presented concurrently as part of a touchscreen control interface of the application;

the touchscreen control interface is configured such that the second selectable control element and the second selectable control element control controls all smart-home devices having the common function of the first smart home device and the second smart home device located within a structure regardless of location within the structure; and

the touchscreen control interface is configured such that the first selectable control element controls only smart-home devices having the common function at the location.

11. The system for integrating control of multiple cloud-based smart-home devices of claim 10, wherein:

a first third-party manufacturer that provides the first smart-home device operates a first cloud-based server system used to control the first smart-home device; and

a second third-party manufacturer that provides the second smart-home device operates a second cloud-based server system used to control the second smart-home device, the second third-party manufacturer being distinct from the first third-party manufacturer and the second cloud-based server system being distinct from the first cloud-based server system.

12. The system for integrating control of multiple cloud-based smart-home devices of claim **11**, wherein the application further causes the one or more processors of the mobile device to:

cause a message to be transmitted to a cloud-based host system for the first smart-home device, wherein the cloud-based host system controls the common function of the first smart-home device via communication with the first cloud-based server system.

13. The system for integrating control of multiple cloud-based smart-home devices of claim **12**, wherein the message transmitted to the cloud-based host system is further for the second smart-home device, wherein the cloud-based host system controls the common function of the second smart-home device via communication with the second cloud-based server system.

14. The system for integrating control of multiple cloud-based smart-home devices of claim **13**, wherein the location is a room within a structure.

15. The system for integrating control of multiple cloud-based smart-home devices of claim **14**, wherein a smart device count is presented concurrently with the first selectable control element, the smart device count indicating a total number of devices controlled by the first selectable control element.

16. The system for integrating control of multiple cloud-based smart-home devices of claim **15**, wherein the second selectable control element controls all smart devices having the common function within the structure regardless of location within the structure.

17. The system for integrating control of multiple cloud-based smart-home devices of claim **15**, wherein the second selectable control element is presented in the application as part of a top interface region.

18. The system for integrating control of multiple cloud-based smart-home devices of claim **10**, wherein the application further causes the one or more processors of the mobile device to:

present, concurrently with the first selectable control element, a third selectable control element and a fourth selectable control element, wherein the third selectable control element controls only the first smart-home device and the fourth selectable control element controls only the second smart-home device.

19. The system for integrating control of multiple cloud-based smart-home devices of claim **18**, wherein the application further causes the one or more processors of the mobile device to:

present, concurrently with the first selectable control element, a state indication for the common operating characteristic group.

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